Theta manifold geometry and signed heat-flow collision control

Drafted for Edward Baker

Research manuscript — 15 October 2026

Abstract

We study the collision detector $E = H^2 + H_x^2$ for the complete theta heat flow through signed jet geometry and arithmetic source identities. Every finite collision forces a negative fixed-endpoint chord of order $\sqrt{E}$. An exponential average removes the endpoint derivative, and cutting off only the endpoint preserves a $\sqrt{E}$ factor in the error. This yields a conditional exclusion criterion with independently chosen spatial lengths and cutoffs, without a prescribed cutoff-growth rate. The normalized coefficient recurrence identifies theta and admits an exact defect-to-observation formula. Positive finite recurrence relaxations have sharp $O(1/N)$ physical stability; fixing their affine slope improves the bound to a Gaussian tail. A finite-depth adjoint current gives a recurrence-adapted endpoint approximation, while its unresolved lower trace explains why recurrence inversion alone supplies no sign. The geometric formulation separates physical orientation from radial vanishing order and gives a weaker signed boundary criterion. Complete source identities retain their phases, transverse jets and primitive endpoints. Positive-source collision controls and coupled-product obstructions are summarized with references to their detailed proofs. The remaining problem is an independent theta-specific signed trace estimate with full domain coverage. No new global theta sign bound or proof of RH is claimed.

Contents

1 The geometric problem and the program **2**
- 1.1 Results and their logical status . . . 3
- 1.2 Relation to the preceding investigation . . . 4
- 1.3 Conventions . . . 4

2 The physical heat field and its collision obstruction **4**
- 2.1 Full stationary predecessors at every finite multiplicity . . . 6

3 Signed forms and the physical observation **7**
- 3.1 The Hodge pairing in four dimensions . . . 7
- 3.2 Primitives and a signed boundary observation . . . 8
- 3.3 An integrated positive identity and its scale . . . 9

4 Hyperkähler lifts and their degenerations **9**
- 4.1 A flat lift with an exact physical coefficient . . . 9
- 4.2 Conformal and compact obstructions . . . 10

5 A holomorphic surface with the physical metric coefficient **11**
5.1 Curvature and induced holonomy . . . 11
5.2 Genuine calibrated inclusions and their limits . . . 12

6 The marked real Gauss map retains the sign **13**
6.1 A sign-reversing geometric control . . . 14

7 Conditional exclusion by signed chords and spatial turning **14**
7.1 A fixed nearby endpoint and starting-point energy . . . 14
7.2 Transfer from a complete family of cutoff estimates . . . 16
7.3 Exponential averages and complete-start cutoff transfer . . . 17
7.4 The inner profile and quarter-turn concentration . . . 19
7.5 A supporting one-sided continuity criterion . . . 20

8 The complete theta source on a curved elliptic family **21**
8.1 The complete primitive and its boundary contribution . . . 21
8.2 The theta line bundle, its connection, and the fourth jet . . . 22
8.3 A nonflat hyperkähler metric with the correct elliptic fibers . . . 24
8.4 An actual abelian anti-self-dual theta connection . . . 26
8.5 Physical restriction and complete signed source identities . . . 26

9 Arithmetic discrimination: support, endpoints and reflection **28**
9.1 A validated collision within the square-support relaxation . . . 29
9.2 Complete reflection fixes the coefficients . . . 30
9.3 A normalized coefficient recurrence . . . 31
9.4 Coupled addition and the complete starting factor . . . 32

10 Quantitative recurrence stability and endpoint currents **33**
10.1 Physical mode bounds and the imaginary anchor . . . 33
10.2 An exact normalized defect identity . . . 33
10.3 A finite positive hierarchy and its sharp rate . . . 34
10.4 A finite current preserving the starting amplitude . . . 35

11 Arithmetic boundary traces and signed observation **36**

12 The remaining arithmetic-to-physical estimate **38**
12.1 A signed trace from a joint source identity . . . 39
12.2 Earlier unsuccessful constructions . . . 39
12.3 Validation and coverage . . . 39

A Verification scope and provenance **40**

1 The geometric problem and the program

The theta heat flow presents a tension between positive source data and a signed spectral observation. The complete source is positive and has very rapid decay, but its oscillatory transform can have small first jets. The physical collision detector is

$$E(t,x) = H(t,x)^2 + H_x(t,x)^2.$$

The research objective is to exclude $E = 0$ at finite positive heat time on the full real spectral line. The established de Bruijn–Newman framework supplies the relation to RH, stated with its domain requirements below. This paper develops the differential geometry of that objective; it does not prove the missing theta-specific estimate.

There are two natural manifolds. On the physical side, a primitive $P_z = H$ gives a holomorphic surface $X_t(z, w) = (P(t, z), H(t, z), w)$ whose induced metric contains the complex first-jet energy as a coefficient. Its regular locus is a Kähler manifold; a common zero of H and H_z is a branch event for the parametrization. The real projective tangent direction retains the signed Laguerre quantity through

$$\partial_x \arg(H + iH_x) = -\frac{H_x^2 - HH_{xx}}{E}.$$

The associated two-point observation is

$$\mathcal{C}_j(t; x, y) = h_j(t, x)\{h_j(t, y) - h_{j+1}(t, y)\}, \qquad h_j = \partial_x^j H, \quad j = 0, 1.$$

At a full stationary start $h_{j+1}(t, x) = 0$, on an interval with $h_j \neq 0$, its nonnegativity is precisely a $\pi/4$ upper bound on the rightward projective turning. More usefully, every collision forces $\mathcal{C}_j/\sqrt{E(t, x)}$ to approach a strictly negative constant at a fixed nearby endpoint. A lower bound with error $o(\sqrt{E(t, x)})$ therefore suffices across all finite multiplicities. The energy is evaluated at the starting point on both parity branches.

On the source side, the actual theta kernel is a covariant derivative expression in a degree-one line bundle over a curved elliptic family. The complete two-source transform retains its spectral phases, positive heat multiplier, transverse jet corrections and primitive endpoints. Its directly restricted connection is flat, so curvature alone adds no new tangential sign. The arithmetic question is sharper: which coupling of the theta coefficients can control this complete signed observation? A positive weighted-square source can collide while retaining the first source endpoint; full modular reflection excludes those free weights by forcing the theta coefficients. Turning this rigidity into an inequality remains open.

1.1 Results and their logical status

The principal results and the remaining candidate are as follows.

1. The fixed-endpoint chord theorem gives conditional collision exclusion from an $o(\sqrt{E})$ starting-energy error at every finite multiplicity. The exponent $1/2$ is sharp for arbitrary power-law allowances; the inner heat profiles resolve the turning concentration.

2. Exponential averaging gives the endpoint observation $W^2 - e^{-\delta}W(x)W(x + \delta)$. Endpoint-only approximation preserves the full starting factor. A mixed bound $-K_N E - \varepsilon_N\sqrt{E}$ suffices with independently chosen cutoffs, even when the locally bounded constants K_N grow arbitrarily with N. The earlier additive-tail transfer remains available under its stated rate bounds.

3. Exact normalized recurrence defects control physical mode observations with Gaussian dual weights. Positive recurrence relaxations through index N have sharp $O(1/N)$ stability, improved to a Gaussian beta tail by a slope anchor. These are stability results, not favorable-sign estimates at the exact theta coefficients.

4. A finite neighboring-mode current gives an explicit $O(L^{-1/2})\sqrt{E}$ endpoint transfer error. Its lower trace retains the unknown observation; the failed normalization cancellation and the finite-source obstruction are stated briefly with note references.

5. Complete signed source formulas and coupled addition retain independent source variables, parity, exchange terms and primitive endpoints. Positive-source controls show why weakened arithmetic hypotheses are insufficient; complete reflection fixes the input.

6. Signed forms, holomorphic jet surfaces, calibrated inclusions and theta connections supply the geometric setting. The recurrence trace has a one-dimensional cokernel, and bounded endpoint descent to the physical-jet quotient requires control through degeneration. A signed trace estimate is weaker than absolute endpoint reconstruction.

The remaining task is an independently proved signed boundary estimate using the complete theta coupling. The analytic collision criteria are conditional; the control sources are not theta; finite checks validate specified calculations. These roles must not be conflated with a new theta sign domain or an RH conclusion. We make no novelty claim for the standard geometric constructions.

1.2 Relation to the preceding investigation

This investigation separates the manifold and connection direction from the broader signed-theta program in [1]. The preceding manuscript develops complete-source correlations, cutoff accounting, outer-region estimates and finite certificates. Its self-dual construction and its later manifold notes are the starting point of the present narrative. The outer estimates and certificates retain their original statements and verification records; they are not reproved or used as a replacement for the missing global geometric estimate here.

The author's exploratory quaternion-form notes of January and February 2014 [2] motivate preserving a signed, oriented pairing. We build the required identities directly and state their boundary hypotheses. The new line-bundle construction is abelian and has an explicit connection and curvature; it invokes no unproved general nonabelian boundary formula.

Throughout, a regular jet interval means an interval on which the particular pair defining the angle is nowhere zero. Source-prescribed constants or moduli must be derived from the complete theta data and remain uniform through neighborhoods of hypothetical collisions. A constant chosen retrospectively from an observed positive energy floor does not provide such an estimate.

1.3 Conventions

All metrics are real Riemannian metrics. We use $|dz|^2 = dx^2 + dy^2$ and the Kähler form $(i/2)\, dz \wedge d\bar{z}$. The physical time t is an external parameter for the four-dimensional exterior derivatives. It is distinct from the elliptic modulus τ, whose imaginary part is positive. Hodge signs refer to the orientation stated in each construction. For a Hermitian holomorphic line bundle in a holomorphic frame with squared norm h, we use the Chern connection $\nabla = d + \partial \log h$ and curvature $F_\nabla = \overline{\partial}\partial \log h$; a frame change transforms the local connection in the usual way.

2 The physical heat field and its collision obstruction

The geometric constructions in this paper must retain a specified physical observation. We first fix that observation and prove the local analytic facts used later. The conclusions in this section about arbitrary analytic heat fields are local theorems. Their implication for the Riemann hypothesis uses an explicitly identified earlier theorem for the complete theta source.

For real $u \geq 0$, define

$$\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{2.1}$$

$$H(t,z)=\int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du. \tag{2.2}$$

The variable t is an external heat parameter. In the geometry below it is fixed when spatial derivatives, metrics, and curvature are computed. On real (t,x) put

$$h_j=\partial_x^j H,\qquad E=h_0^2+h_1^2,\qquad \mathcal{L}_j=h_{j+1}^2-h_jh_{j+2},\qquad S_j=\{h_{j+1}=0\},\quad j=0,1. \tag{2.3}$$

The same notation $h_j(t,z)$ denotes the holomorphic spectral derivatives. The four consecutive components h_0,h_1,h_2,h_3 have one scalar source; they are not independent fields to which arbitrary geometric constraints may be assigned.

Lemma 2.1 (Analyticity and the sign of heat time)**.** *The integral (2.2) defines a jointly entire function of $(t,z)\in\mathbb{C}^2$. It is even in z, obeys real reflection for real t, and satisfies*

$$H_t=-H_{zz}. \tag{2.4}$$

For every real t, $H(t,0)>0$. Consequently no real-time slice is identically zero and every spatial zero has finite multiplicity.

Proof. Every term of (2.1) is positive for $u\ge 0$, since $2\pi n^2e^{4u}-3>0$. Dropping the negative term and comparing successive terms of $n^4e^{-\pi n^2e^{4u}}$ gives

$$0<\Phi(u)<4\pi^2e^{9u-\pi e^{4u}}.$$

Indeed the ratio of successive majorant terms is at most $16e^{-3\pi}<1/2$. On each compact set of complex (t,z), every fixed parameter derivative of the integrand is bounded by a constant times

$$(1+u)^M\exp\{Cu^2+Cu+9u-\pi e^{4u}\},$$

which is integrable. Uniform convergence and differentiation under the integral prove joint entirety and (2.4). The symmetries follow from the cosine representation. At real t and $z=0$ the integrand is strictly positive. Finally, zeros of a nonzero holomorphic function have finite order. □

Direct differentiation, with x held fixed, gives

$$\frac{1}{2}E_t=-h_0h_2-h_1h_3=\begin{cases}\mathcal{L}_0, & h_1=0,\\ \mathcal{L}_1, & h_2=0.\end{cases} \tag{2.5}$$

These are the stationary sets of the original field and its first derivative. A normalization of the field depending on x would change its stationary set and cannot be substituted in this identity.

Remark 2.2 (The imported theta-to-RH bridge). For the normalization (2.1)–(2.2), the earlier investigation establishes

$$H(0,z)=\frac{1}{8}\xi\left(\frac{1}{2}+\frac{iz}{2}\right)$$

and the finite-collision criterion

$$\text{RH}\quad\Longleftrightarrow\quad E(t,x)>0\quad(0<t\le 1/2,\ x\in\mathbb{R}). \tag{2.6}$$

We use this result as an imported theorem, with its normalization and range unchanged; see [1, section on positive-time collisions]. Its global inputs are the de Bruijn–Newman real-zero threshold, including its endpoint and upper bound $1/2$, the strip bound for nonreal zeros, and a confinement estimate uniform for t in compact positive-time intervals. The last input prevents a limiting first collision from escaping to infinite real height. Joint analyticity and conjugation symmetry then force a finite multiple real zero if RH fails. Conversely, local backward-heat splitting at a positive-time multiple zero produces nonreal zeros immediately before it. Thus a geometric construction alone does not supply (2.6); its global theta hypotheses remain part of the imported theorem. The local exclusion arguments below are proved independently of that import.

2.1 Full stationary predecessors at every finite multiplicity

Lemma 2.3 (Even-order stationary predecessor)**.** *Let $W(t,x)$ be real analytic near (T,a), satisfy $W_t = -W_{xx}$, and have an exact spatial zero of order $2r$ there, where $r \geq 1$. Write $A = \partial_x^{2r} W(T,a) \neq 0$ and $B = \partial_x^{2r+1} W(T,a)$. For sufficiently small $s \geq 0$ there is a real analytic curve $\chi(s)$ such that*

$$W_x(T-s, \chi(s)) = 0, \qquad \chi(s) = a - \frac{B}{rA} s + O(s^2), \tag{2.7}$$

$$W(T-s, \chi(s)) = \frac{A}{r!} s^r + O(s^{r+1}), \qquad W_{xx}(T-s, \chi(s)) = \frac{A}{(r-1)!} s^{r-1} + O(s^r). \tag{2.8}$$

In particular the central value of W is nonzero for sufficiently small $s > 0$.

Proof. Analyticity and the heat equation give the convergent Taylor expansion

$$W(T-s, a+y) = \sum_{p,q \geq 0} \frac{s^p y^q}{p! q!} \partial_x^{2p+q} W(T,a). \tag{2.9}$$

Set $y = s\eta$ and differentiate once in y. Every surviving term has $2p + q + 1 \geq 2r$, hence $p + q \geq r$. It follows that $s^{-r} W_x(T-s, a+s\eta)$ extends analytically across $s = 0$. At $s = 0$ its value is

$$\frac{A}{(r-1)!} \eta + \frac{B}{r!}.$$

Its derivative in η is nonzero. The analytic implicit-function theorem supplies $\eta(s)$ with initial value $-B/(rA)$, proving (2.7). Substitution in (2.9) proves (2.8): the leading terms are respectively $p = r, q = 0$ and, after two spatial derivatives, $p = r-1, q = 0$. □

Proposition 2.4 (The physical stationary pole)**.** *Suppose a nonzero analytic backward-heat field H has exact spatial multiplicity $m = 2r + j$ at (T,a), with $r \geq 1$ and $j \in \{0,1\}$. Apply Lemma 2.3 to $W = h_j$. On its full stationary predecessor, as $s \downarrow 0$,*

$$h_{j+1} = 0, \qquad E > 0, \qquad \frac{\mathcal{L}_j}{E} = -\frac{r}{s} + O(1). \tag{2.10}$$

For $j = 0$ one has $E = W^2$ exactly. For $j = 1$,

$$H(T-s, \chi(s)) = O(s^{r+1}), \qquad E = W^2(1 + O(s^2)). \tag{2.11}$$

Proof. On the constructed branch $\mathcal{L}_j = -WW_{xx}$. Equation (2.8) gives $W_{xx}/W = r/s + O(1)$. If $j = 0$, stationarity is $H_x = 0$, so $E = W^2$ and the claim follows. If $j = 1$, H has order $2r+1$ at the event. In its heat Taylor expansion evaluated at $x - a = O(s)$, each surviving term has $2p + q \geq 2r + 1$, which implies $p + q \geq r + 1$. Thus $H = O(s^{r+1})$, whereas W is a nonzero constant times s^r. This proves (2.11) and then (2.10). All divisions are taken on the proved nonzero predecessor, not at the event itself. □

Theorem 2.5 (Signed stationary exclusion)**.** *Let H be a real analytic backward-heat field whose spatial slices are not identically zero. Suppose nonnegative functions K_0, K_1 are locally bounded through every candidate event in a stated time-space domain and*

$$\mathcal{L}_j \geq -K_j E \quad \text{on the full set } S_j, \qquad j = 0, 1. \tag{2.12}$$

Assume the domain contains the sufficiently close predecessors of each candidate event. Then it contains no multiple real zero covered in this way. For the complete theta field, if the hypothesis covers $(0, 1/2] \times \mathbb{R}$, it implies RH by Remark 2.2.

Proof. Every multiple zero has a finite order $m \geq 2$, uniquely expressible as $2r + j$ with the stated ranges. Proposition 2.4 then contradicts the locally bounded coefficient K_j on its corresponding predecessor. At the endpoint $t = 1/2$, a left-hand time neighborhood suffices. No boundedness as $t \downarrow 0$ or as $|x| \to \infty$ is required. □

Both branches are essential for this argument. The backward-heat polynomial $H(t, x) = x^3 - 6(t - T)x$ has $\mathcal{L}_0 = 3x^4 + 36(t - T)^2 \geq 0$ everywhere and nevertheless has a triple zero at $(T, 0)$. On the second stationary predecessor $x = 0$, $\mathcal{L}_1/E = -1/(T - t)$.

3 Signed forms and the physical observation

The two Laguerre expressions are polarized quadratic forms in three consecutive derivatives. A geometric representation must preserve that polarization. Replacing it by the norm of the whole jet changes the question, because the higher derivatives need not vanish when E does. The quaternion-form working notes of Edward B. Baker III [2] motivate the use of oriented two-form pairings. The construction below is derived directly; it uses none of the unfinished singular-kernel or nonabelian boundary formulas in those notes.

3.1 The Hodge pairing in four dimensions

Fix the Euclidean metric on $\mathbb{R}^4$ with coordinates (x, y, a, b), orientation $dx \wedge dy \wedge da \wedge db$, and complex coordinates $z = x + iy$, $w = a + ib$. This use of a, b avoids confusing the auxiliary manifold coordinates with the source variable u. The Hodge star gives

$$\Lambda^2 = \Lambda^2_+ \oplus \Lambda^2_-, \qquad \alpha \wedge \alpha = (|\alpha_+|^2 - |\alpha_-|^2)\, dV.$$

The wedge pairing has signature $(3, 3)$; the inner product $\alpha \wedge *\alpha$ is positive definite. Both structures are useful, but they answer different questions.

For orthonormal forms $\sigma_1^+, \sigma_2^+ \in \Lambda^2_+$ and $\sigma_1^- \in \Lambda^2_-$, the form

$$\Omega = B\sigma_1^+ + \tfrac{1}{2}(A - C)\sigma_2^+ + \tfrac{1}{2}(A + C)\sigma_1^-$$

has wedge square $(B^2 - AC)dV$ and squared norm $B^2 + (A^2 + C^2)/2$. This realizes a Laguerre expression algebraically, but its norm pays for the higher jet C, and a variable-coefficient version need not be closed. We instead realize each input by a closed form and retain the signed polarization explicitly.

Proposition 3.1 (Closed self-dual inputs)**.** *For a holomorphic function $f(z)$, define*

$$\alpha_f = \frac{1}{\sqrt{2}} \operatorname{Re}(f(z)\, dz \wedge dw).$$

Then

$$d\alpha_f = 0, \qquad *\alpha_f = \alpha_f, \qquad \alpha_f \wedge \alpha_g = \operatorname{Re}(f\overline{g})\, dV. \tag{3.1}$$

The form is Hodge-harmonic, and in the fixed Euclidean metric

$$|\alpha_f|^2 = |f|^2, \qquad |\nabla \alpha_f|^2 = 2|f'|^2. \tag{3.2}$$

In particular, it is parallel on a connected domain exactly when f is constant.

Proof. Put $A = dx \wedge da - dy \wedge db$ and $B = dx \wedge db + dy \wedge da$. Both are self-dual, $A^2 = B^2 = 2dV$, and $A \wedge B = 0$. If $f = p + iq$, then $\alpha_f = (pA - qB)/\sqrt{2}$, which proves the pairing and norm. Holomorphicity gives $d(f\,dz \wedge dw) = 0$. Self-duality then implies $d^*\alpha_f = - * d * \alpha_f = 0$. The Cauchy–Riemann equations give $|\nabla p|^2 + |\nabla q|^2 = 2|f'|^2$, proving the last identity. □

For $f = h_j(t, \cdot)$, heat time is held fixed in this exterior derivative. Writing $\alpha_j = \alpha_{h_j}$, we obtain

$$\alpha_{j+1} \wedge \alpha_{j+1} - \alpha_j \wedge \alpha_{j+2} = \big(|h_{j+1}|^2 - \mathrm{Re}(h_j \overline{h_{j+2}})\big) dV. \tag{3.3}$$

On $y = 0$ the coefficient is $\mathcal{L}_j$. This is evaluation of a four-form coefficient on the real slice, not pullback of a four-form onto a one-dimensional line. The coefficient-space matrix has spectrum $\{1, 1/2, -1/2\}$. Its negative eigenvalue persists even though all three input forms are closed and self-dual.

3.2 Primitives and a signed boundary observation

If $P_f' = f$, then

$$\beta_f = \frac{1}{\sqrt{2}} \mathrm{Re}(P_f\, dw), \qquad d\beta_f = \alpha_f.$$

Stokes' theorem gives, for a smooth bounded four-domain with the forms defined on a neighborhood of its closure,

$$\int_\Omega \alpha_f \wedge \alpha_g = \int_{\partial\Omega} \beta_f \wedge \alpha_g. \tag{3.4}$$

This retains a mixed signed pairing. It does not replace the boundary term by a positive norm or remove it.

The following elementary identity makes the same distinction on a spectral circle. Write $\langle q \rangle = (2\pi)^{-1} \int_0^{2\pi} q(\theta)\, d\theta$.

Proposition 3.2 (Complete circle current)**.** *Let f be holomorphic near the closed disk $|z - x_0| \le R$, real on its real diameter, and $f'(x_0) = 0$. Set*

$$\varphi(\theta) = f(x_0)\, \mathrm{Re}\, f(x_0 + Re^{i\theta}).$$

Then

$$-f(x_0) f''(x_0) = -\frac{4 f(x_0)^2}{R^2} + \frac{4}{R^2} \langle (1 - \cos 2\theta) \varphi(\theta) \rangle. \tag{3.5}$$

For $f = h_j$ this is exactly $\mathcal{L}_j$ on its full stationary set.

Proof. The Taylor coefficients are real. The mean of $\mathrm{Re}\, f$ is $f(x_0)$ and its second cosine coefficient has mean pairing $\langle \cos 2\theta\, \mathrm{Re}\, f \rangle = R^2 f''(x_0)/4$. Substitution proves the formula. □

The current φ is also the coefficient of $\alpha_{f(x_0)} \wedge \alpha_f$ evaluated on the circle. A useful bound must control the complete weighted current relative to E, with all radius costs accounted for. Separately estimating its negative mass can discard essential favorable cancellation.

3.3 An integrated positive identity and its scale

The following result, inherited from the signed-form development in [1], explains what positivity of a completed integral can provide without supplying the central pointwise estimate.

Theorem 3.3 (Disk completion)**.** *For f holomorphic near $\overline{D_R(x_0)}$, define*

$$\ell_f = |f'|^2 - \mathrm{Re}(f\overline{f''}), \qquad g_R(z) = \left(1 - \frac{(z-x_0)^2}{R^2}\right) f(z).$$

Then

$$\int_{D_R} \ell_f \, dA + \frac{1}{2R}\int_{\partial D_R} |f|^2 \, ds = \frac{1}{2}\int_{D_R} |g_R'|^2 \, dA \geq 0. \tag{3.6}$$

The boundary coefficient cannot be decreased, even for real-reflection polynomials stationary at the center.

Proof. Write $f(x_0 + R\zeta) = \sum_{n\geq 0} b_n \zeta^n$ and set $b_n = 0$ for $n < 0$. Angular orthogonality gives

$$\begin{aligned}
\pi^{-1}\int_{D_R} \ell_f \, dA &= \sum_{n\geq 0} n|b_n|^2 - \mathrm{Re}\sum_{n\geq 0}(n+2)b_n\overline{b_{n+2}},\\
(2\pi R)^{-1}\int_{\partial D_R} |f|^2 \, ds &= \sum_{n\geq 0} |b_n|^2,\\
\pi^{-1}\int_{D_R} |g_R'|^2 \, dA &= \sum_{n\geq 1} n|b_n - b_{n-2}|^2.
\end{aligned}$$

Expansion proves (3.6). For $f_N = \sum_{m=0}^{N}(1 - m/(N+1))((z-x_0)/R)^{2m}$, direct summation gives

$$B_N := \frac{1}{2R}\int_{\partial D_R} |f_N|^2 \, ds = \pi\frac{(N+2)(2N+3)}{6(N+1)}, \qquad \frac{\int_{D_R}\ell_{f_N}\,dA}{B_N} = -\frac{2N}{2N+3} \longrightarrow -1.$$

These polynomials have $f_N(x_0) = 1$ and $f_N'(x_0) = 0$. □

Their boundary energy grows while their central first-jet energy stays equal to one. Thus the integrated positive completion cannot be converted to a uniform central-energy payment for arbitrary holomorphic fields. The actual theta source must supply the additional restriction.

4 Hyperkähler lifts and their degenerations

A hyperkähler metric in real dimension four has a parallel compatible triple of complex structures satisfying the quaternion relations; its holonomy is contained in $\mathrm{Sp}(1) = \mathrm{SU}(2)$. In particular it is Ricci-flat. These are statements about the metric and its parallel tensors, not a consequence of closed self-duality of one form; see [4].

4.1 A flat lift with an exact physical coefficient

Proposition 4.1 (Pullback hyperkähler metric)**.** *Where a holomorphic $f(z)$ is nonzero, the metric*

$$g_f = |f(z)|^2|dz|^2 + |dw|^2 \tag{4.1}$$

is locally flat and hyperkähler. Its parallel triple is

$$\omega_1 = \frac{i}{2}(|f|^2 dz \wedge d\overline{z} + dw \wedge d\overline{w}),$$
$$\omega_2 = \mathrm{Re}(f\, dz \wedge dw), \qquad \omega_3 = \mathrm{Im}(f\, dz \wedge dw).$$

For $f = H + iH_z$, its real-slice coefficient is exactly E.

Proof. A local primitive $F' = f$ supplies the nonsingular holomorphic coordinate $\zeta = F(z)$. Then $g_f = |d\zeta|^2 + |dw|^2$ and the displayed forms are the standard constant hyperkähler triple in those coordinates. Equivalently they are closed and $\omega_i \wedge \omega_j = 2\delta_{ij}|f|^2 dV$. For real x, both H and H_x are real, so $|H + iH_x|^2 = H^2 + H_x^2$. □

This construction has trivial restricted holonomy, contained in Sp(1); it does not have holonomy equal to the full group. It is a precise geometric realization of the physical scale. At a zero of f of order m, however, $F - F(z_0)$ has order $m+1$ and (4.1) degenerates. After a nonsingular holomorphic change of coordinate its curve factor is a constant times $|\zeta|^{2m}|d\zeta|^2$. Its metric completion has cone angle $2\pi(m+1)$ and finite distance to the vertex. Passing to the branched coordinate F is not a nonsingular coordinate repair on the original domain.

At a nondegenerate physical collision $H = H_z = 0$, $H_{zz} \neq 0$, the coefficient $f = H + iH_z$ has a simple zero since $f' = iH_{zz}$. Thus the real physical collision produces an angle-4π degeneration. Away from the real slice, $H + iH_z$ can vanish without both jets vanishing; the exact equivalence with the physical detector is a real-slice statement. The fixed background metric remains smooth at every such event, but its theta-dependent forms are generally harmonic rather than parallel.

4.2 Conformal and compact obstructions

In dimension four the Hodge star on two-forms is unchanged by $g \mapsto e^{2\sigma} g$. Thus closed self-dual forms remain harmonic under arbitrary conformal changes. For example, $e^{2x} g_0$ has

$$\mathrm{Ric} = -2(dy^2 + da^2 + db^2), \qquad \mathrm{Scal} = -6e^{-2x}.$$

It is not hyperkähler, despite preserving all the closed self-dual inputs. The top-form pairing itself is unchanged; only its coefficient relative to the volume changes, by a positive factor $e^{-4\sigma}$. No sign improvement follows from this conformal replacement.

Pointwise normalization also needs care. Where $f \neq 0$,

$$d(\alpha_f/|f|) = -d\log|f| \wedge (\alpha_f/|f|).$$

A nonzero self-dual two-form is symplectic, so wedge multiplication by it is an isomorphism from one-forms to three-forms. Consequently this normalized form is closed only if $|f|$ is constant, which for holomorphic f forces f constant. Normalization can therefore lose the differential equation needed for a holonomy construction.

Proposition 4.2 (Compact exactness obstruction)**.** *On a compact oriented four-manifold without boundary, a globally exact self-dual two-form vanishes.*

Proof. If $\alpha = d\beta$ and $*\alpha = \alpha$, then $\int |\alpha|^2 dV = \int \alpha \wedge \alpha = \int d(\beta \wedge \alpha) = 0$. □

Our forms are exact on $\mathbb{C}^2$. A nontrivial compact realization must therefore change their global period, boundary or singularity data; a descended form may have a primitive that fails to descend. On the original noncompact space, the forms are independent of w and are not square integrable

unless zero. Compact integration by parts cannot be imported without controlling the missing boundaries.

There is a further restriction on extending the entire input family to a compact hyperkähler manifold. With the orientation in which the parallel triple is self-dual, the Weitzenböck formula for a harmonic self-dual form is

$$0 = \nabla^*\nabla\alpha + (\mathrm{Scal}\,/3 - 2W^+)\alpha = \nabla^*\nabla\alpha.$$

Here scalar curvature and W^+ vanish. Integration implies parallelness. This makes metric-relative pairings constant, although coordinate coefficients of a parallel form need not be constant. For the actual family a stronger local argument applies: at a real point with $H \neq 0$, the forms α_H and α_{H_z} are proportional with a real ratio c. If both were parallel for the same connection, their difference $\alpha_{H_z} - c\alpha_H$ would vanish throughout the connected patch. This forces $H_z = cH$, incompatible with a nonzero even entire field decaying along the real axis. This obstruction concerns preservation of the whole particular family, not every possible compact geometric representation.

5 A holomorphic surface with the physical metric coefficient

Fix real t, abbreviate $f(z) = H(t, z)$, and choose a holomorphic primitive $P' = f$. For the entire theta field a global primitive exists. Define

$$\mathcal{E}(z) = |f(z)|^2 + |f'(z)|^2, \qquad \Omega_t = \{z : \mathcal{E}(z) > 0\}.$$

On $\Omega_t \times \mathbb{C}$ consider

$$X_t(z, w) = (P(z), f(z), w) \in \mathbb{C}^3. \tag{5.1}$$

The complex tangent vectors $(f, f', 0)$ and $(0, 0, 1)$ are independent exactly where $\mathcal{E} > 0$. Thus (5.1) is a holomorphic immersion with induced real metric and Kähler form

$$g_t = \mathcal{E}(z)|dz|^2 + |dw|^2, \tag{5.2}$$

$$\omega_t = \frac{i}{2}\{\mathcal{E}\, dz \wedge d\bar{z} + dw \wedge d\bar{w}\}. \tag{5.3}$$

The parameter domain with this metric is a smooth Riemannian four-manifold. Its image is an immersed complex surface; global injectivity and absence of self-intersections are not asserted. On the physical real slice, $\mathcal{E}(x) = E(t, x)$ exactly. At a common zero of H and H_z, the map loses one complex rank and (5.2) is not positive definite.

5.1 Curvature and induced holonomy

Proposition 5.1 (Curvature of the jet surface)**.** *Set $D = ff'' - (f')^2$. With the real metric convention $|dz|^2 = dx^2 + dy^2$, the Gaussian curvature of the first factor of (5.2) is*

$$K_t = -\frac{2|D|^2}{\mathcal{E}^3}. \tag{5.4}$$

The scalar curvature of the four-dimensional product is $2K_t$. Its Ricci tensor equals K_t times the curve metric in the first factor and zero in the flat factor; its Ricci form is

$$\rho_t = -i\frac{|D|^2}{\mathcal{E}^2}\, dz \wedge d\bar{z}. \tag{5.5}$$

In particular, on the physical line,

$$K_t(x) = -\frac{2\mathcal{L}_0(t,x)^2}{E(t,x)^3}. \tag{5.6}$$

Proof. The two-by-two Hermitian Gram determinant gives

$$\begin{aligned}\partial_z\partial_{\bar z}\log\mathcal{E} &= \frac{\mathcal{E}(|f'|^2+|f''|^2)-|f'\bar f+f''\overline{f'}|^2}{\mathcal{E}^2}\\ &= \frac{|ff''-(f')^2|^2}{\mathcal{E}^2}.\end{aligned}$$

For $ds^2=\mathcal{E}(dx^2+dy^2)$ the Gaussian curvature is $-(2\mathcal{E})^{-1}\Delta\log\mathcal{E}$, and $\Delta=4\partial_z\partial_{\bar z}$. This proves (5.4). Product curvature proves the Ricci tensor and scalar statements. Equivalently the Kähler Ricci form is $-i\partial\bar\partial\log\det(g_{a\bar b})$; the irrelevant constant in the Hermitian determinant disappears upon differentiation, giving (5.5). On the real line $D=-\mathcal{L}_0$, proving the last identity. □

The metric has a parallel splitting into a curve and a flat complex line, so its holonomy is contained in $U(1)\times\{1\}\subset U(2)$. It is Ricci-flat on a connected patch precisely when $D\equiv 0$ there. That condition forces

$$f(z)=Ae^{bz}, \qquad A\neq 0, \tag{5.7}$$

on that patch. Indeed on a subpatch where $f\neq 0$ one has $(f'/f)'=D/f^2=0$, and holomorphic identity continuation proves (5.7). If this equality held on an open patch of a theta slice, the identity theorem would extend it to the whole spectral plane. A nonzero even theta field cannot be such an exponential: evenness would force $b=0$, whereas $H_{zz}(t,0)=-\int_0^\infty u^2e^{tu^2}\Phi(u)\,du<0$. Thus this induced metric is not a hyperkähler metric for the theta field. This is a statement about (5.2), not a prohibition on other auxiliary metrics or other geometric realizations.

The squaring in (5.6) is decisive. A lower or upper curvature norm does not select the favorable sign of $\mathcal{L}_0$. In addition, the factor E^{-3} makes intrinsic curvature very sensitive to degeneration of the physical metric.

5.2 Genuine calibrated inclusions and their limits

Every locally embedded image patch of (5.1) is calibrated by the ambient Kähler calibration $\omega_0^2/2$ and hence is minimal. The relevant algebraic and variational statements are the complex submanifold calibration and the fundamental theorem of calibrations; see [5, Theorems 3.1 and 2.3]. This conclusion uses a proved holomorphic immersion, rather than an analogy between quaternionic coefficients and tangent vectors.

There are literal exceptional-calibration realizations as well. On $\mathbb{R}_r\times\mathbb{C}^3$ use the standard G_2 form

$$\varphi_0 = dr\wedge\omega_0+\operatorname{Re}\Omega_3, \qquad *_0\varphi_0=\tfrac12\omega_0^2-dr\wedge\operatorname{Im}\Omega_3.$$

The image of (5.1) in $\{0\}\times\mathbb{C}^3$ is coassociative: the second term vanishes and the first restricts to its induced volume. Alternatively, regard $(P,f,w,0)$ as a surface in $\mathbb{C}^4\cong\mathbb{R}^8$. For the standard Cayley calibration

$$\Psi_0=\tfrac12\omega_0^2+\operatorname{Re}\Omega_4, \tag{5.8}$$

the holomorphic four-form pulls back to zero, while the first term is the volume. Hence this image is Cayley. These standard inclusions are also recorded in [5, Propositions 6.5 and 7.3]. The ambient metrics used here are flat, with trivial holonomy contained in the indicated groups. We have not produced an ambient metric whose holonomy is exactly G_2 or Spin(7).

Calibration does not remove a branch degeneration. If at a fixed time

$$f(z) = c(z-a)^m + O((z-a)^{m+1}), \qquad c \neq 0, \quad m \geq 2,$$

then

$$\mathcal{E}(z) \sim m^2|c|^2|z-a|^{2m-2}. \tag{5.9}$$

The radial distance to a is asymptotic to $|c||z-a|^m$, and the circumference of a small coordinate circle is asymptotic to $2\pi m|c||z-a|^m$. Thus the completion has asymptotic cone angle $2\pi m$ in the curve factor. It is at finite distance but is not a nondegenerate smooth point of the original metric. One can instead make the everywhere immersive graph $(z, P, H, w) \in \mathbb{C}^4$. Its induced metric is $(1+\mathcal{E})|dz|^2 + |dw|^2$, and it remains a calibrated complex surface through a physical joint zero. This repairs immersion by introducing a background metric term; it does not prove nonvanishing of E.

6 The marked real Gauss map retains the sign

The nonconstant tangent direction of (5.1) is the projective Gauss map

$$G_0(t,z) = [H(t,z) : H_z(t,z)] \in \mathbb{CP}^1. \tag{6.1}$$

The second tangent direction is fixed, so this projective line captures the varying part of the full tangent plane. Normalize the Fubini–Study metric by $ds^2_{\mathrm{FS}} = |d\zeta|^2/(1+|\zeta|^2)^2$. Differentiating in either projective chart gives the coordinate-independent identities

$$G_0^* ds^2_{\mathrm{FS}} = \frac{|D|^2}{\mathcal{E}^2}|dz|^2, \qquad \rho_t = -2G_0^*\omega_{\mathrm{FS}}. \tag{6.2}$$

Real reflection puts the physical image in $\mathbb{RP}^1$. Its orientation along the marked real spectral line is extra information which the nonnegative metric in (6.2) omits.

For both branches put

$$E_j^{\mathrm{jet}} = h_j^2 + h_{j+1}^2, \qquad G_j = [h_j : h_{j+1}], \qquad j = 0, 1. \tag{6.3}$$

On a real interval on which this pair never vanishes, choose a continuous real lift $\alpha_j = \arg(h_j + ih_{j+1})$. The lift is defined up to a constant multiple of 2π; its projective value is defined modulo π. Its derivative and every continuously lifted increment are unambiguous. Differentiating the argument proves

$$\boxed{\partial_x \alpha_j = \frac{h_j h_{j+2} - h_{j+1}^2}{h_j^2 + h_{j+1}^2} = -\frac{\mathcal{L}_j}{E_j^{\mathrm{jet}}}.} \tag{6.4}$$

For $j = 0$, $E_0^{\mathrm{jet}} = E$. For $j = 1$ the denominator is different and can vanish even when $E > 0$. No global regularity of G_1 as a physical unit vector is being presumed.

Proposition 6.1 (An oriented angular certificate)**.** *Suppose prescribed functions $\kappa_j \geq 0$ are locally bounded through the covered candidate events and*

$$\partial_x \alpha_j \leq \kappa_j \quad \text{on } S_j \cap \{h_j \neq 0\}, \qquad j = 0, 1.$$

Then $\mathcal{L}_j \geq -\kappa_j E$ on the full stationary sets. Consequently the hypotheses of Theorem 2.5 are satisfied on the same covered domain.

Proof. On S_j with $h_j \neq 0$, (6.4) gives $\mathcal{L}_j \geq -\kappa_j h_j^2 \geq -\kappa_j E$. The second inequality uses both $h_j^2 \leq E$ for $j = 0, 1$ and $\kappa_j \geq 0$. At a remaining point of S_j one has $h_j = 0$, so $\mathcal{L}_j = 0$ and the same inequality holds without defining an angle there. □

For $j = 0$ this certificate is the original signed stationary bound. For $j = 1$ it is in general a stronger sufficient bound, since it pays by h_1^2 on S_1 instead of E. On an odd collision predecessor their ratio tends to one by (2.11). These distinctions are necessary when comparing an auxiliary projective metric with the physical two-component energy.

6.1 A sign-reversing geometric control

Fix $\varepsilon > 0$ and let

$$f_+(z) = \varepsilon + z^2, \quad f_-(z) = \varepsilon - z^2, \qquad P_\pm(z) = \varepsilon z \pm z^3/3.$$

At the real center,

$$E_\pm = \varepsilon^2, \quad \mathcal{L}_{0,+} = -2\varepsilon, \quad \mathcal{L}_{0,-} = 2\varepsilon, \quad K_+ = K_- = -8/\varepsilon^4. \tag{6.5}$$

The corresponding complex surfaces are unitarily congruent after rotating the parameter:

$$\operatorname{diag}(-i, 1, 1)\, X_+(iz, w) = X_-(z, w). \tag{6.6}$$

This follows by substituting iz into each of the two polynomial components. Their intrinsic and calibrated geometry therefore cannot distinguish the two signs without retaining the marked real slice and its orientation. Each polynomial has an exact backward-heat extension $f_\pm(z) \mp 2(t - t_0)$ from its displayed time slice. In particular, $H(t, z) = z^2 - 2(t - T)$ is calibrated at every regular point of its associated surface but develops a double real zero at $(T, 0)$. Its predecessor at $x = 0$ has $\alpha_0' = 1/(T - t)$.

7 Conditional exclusion by signed chords and spatial turning

The marked angle in (6.4) retains a sign that curvature magnitude discards. Its most useful finite-interval observation is a signed chord paid by the physical energy at the stationary starting point. We first prove a multiplicity-independent collision obstruction for that observation, then give the finer turning profiles and the supporting continuity criterion. These are local heat-flow results; the complete-theta estimate required to apply them remains open. The detailed derivations and finite controls are recorded in [9, 10].

7.1 A fixed nearby endpoint and starting-point energy

Put $W = h_j$, $j \in \{0, 1\}$. When $W_x(t, x) = 0$ and W is nonzero on the whole interval $[x, y]$, $y > x$, a continuous angle lift satisfies

$$\alpha_j(t, y) - \alpha_j(t, x) = \arctan \frac{W_x(t, y)}{W(t, y)}. \tag{7.1}$$

Indeed, W has one sign on the interval, so its lifted argument stays in one open half-plane, with no winding ambiguity. In particular,

$$\begin{aligned} \mathcal{C}_j(t; x, y) &:= h_j(t, x)\{h_j(t, y) - h_{j+1}(t, y)\}, \\ \mathcal{C}_j(t; x, y) \geq 0 \quad &\Longleftrightarrow \quad \alpha_j(t, y) - \alpha_j(t, x) \leq \frac{\pi}{4}. \end{aligned} \tag{7.2}$$

All chord losses below use $e = E(t, x)$ at the starting point.

Theorem 7.1 (The fixed-endpoint collision law)**.** *Suppose H has exact spatial multiplicity $2r+j$ at (T,a), $r\ge 1$, and let $W=h_j$, $\chi(s)$ be its full stationary predecessor, and $k=\partial_x^{2r}W(T,a)/r!\neq 0$. For every sufficiently small fixed $\delta>0$, the entire interval $[\chi(s),a+\delta]$ is W-zero-free for sufficiently small $s>0$, and*

$$\boxed{\frac{\mathcal{C}_j(T-s;\chi(s),a+\delta)}{\sqrt{E(T-s,\chi(s))}}\longrightarrow -d_\delta<0,} \qquad d_\delta:=-\operatorname{sgn}(k)\{W(T,a+\delta)-W_x(T,a+\delta)\}>0. \tag{7.3}$$

Proof. Write $a_0=\partial_x^{2r}W(T,a)/(2r)!$. The heat Taylor expansion (2.9), applied to the analytic function $W(T-\eta^2,a+z)$, gives the ordinary two-variable remainder

$$W(T-\eta^2,a+z)=a_0P_{2r}(\eta^2,z)+O((|\eta|+|z|)^{2r+1}), \tag{7.4}$$

$$P_{2r}(\eta^2,z)=\sum_{p=0}^{r}\frac{(2r)!}{p!(2r-2p)!}\eta^{2p}z^{2r-2p},$$

$$P_{2r}(\eta^2,z)\ge(\eta^2+z^2)^r\ge 2^{-r}(|\eta|+|z|)^{2r}. \tag{7.5}$$

For the first inequality, the ratio of the coefficient indexed by p to the corresponding binomial coefficient is $\prod_{i=0}^{p-1}2(2r-2i-1)\ge 1$, with empty product one. Thus the remainder divided by P_{2r} is $O(|\eta|+|z|)$. In a sufficiently small fixed neighborhood, W has the sign of a_0 whenever $(\eta,z)\neq(0,0)$. Since $\chi(s)-a=O(s)$, this proves zero-freeness on the whole asserted interval, not just at its endpoints.

At time T,

$$W(T,a+\delta)-W_x(T,a+\delta)=-2ra_0\delta^{2r-1}+O(\delta^{2r}),$$

which has the opposite sign from k for every sufficiently small positive δ. Finally, (2.8) and (2.11) give

$$W(T-s,\chi(s))=ks^r(1+O(s)),\qquad E(T-s,\chi(s))=k^2s^{2r}(1+O(s)). \tag{7.6}$$

Consequently $W(T-s,\chi(s))/\sqrt{E(T-s,\chi(s))}\to\operatorname{sgn}(k)$; analyticity at the fixed endpoint proves (7.3). □

Theorem 7.2 (Starting-energy chord exclusion)**.** *Let H be a real analytic backward-heat field with no identically zero spatial slice. Near each covered candidate event suppose there are a fixed length $\ell_0>0$, $e_0>0$, and prescribed nonnegative functions $\omega_j(e)=o(\sqrt{e})$ as $e\downarrow 0$, such that*

$$\mathcal{C}_j(t;x,y)\ge-\omega_j(E(t,x)),\qquad j=0,1, \tag{7.7}$$

uniformly whenever $h_{j+1}(t,x)=0$, $0<y-x\le\ell_0$, $0<E(t,x)\le e_0$, and h_j is nonzero on the whole segment, which must lie in the covered neighborhood. Assume that neighborhood contains the sufficiently close predecessors and these short intervals. Then no covered finite multiple real zero occurs. In particular, a bound $\mathcal{C}_j\ge-K_jE(t,x)^\gamma$ suffices for any fixed $\gamma>1/2$, with $K_j\ge 0$ locally bounded uniformly in both endpoints. For the complete theta field, coverage of $0<T\le 1/2$ implies RH by Remark 2.2.

Proof. At a putative event use its parity branch and choose $\delta>0$ in Theorem 7.1, smaller than $\ell_0/2$ and inside the covered neighborhood. The lengths $a+\delta-\chi(s)$ are positive and smaller than ℓ_0 for small $s>0$. Divide (7.7) by the square root of the starting energy and use (7.6). Its lower bound tends to zero, contradicting the strictly negative limit (7.3). At a terminal time a left-hand time neighborhood suffices. □

The fixed positive allowable length and uniformity through the event are essential hypotheses. A bound available only on one prescribed shrinking family need not cover the fixed nearby endpoint used here. Both branches are required because the event multiplicity can have either parity.

Remark 7.3 (Comparison with stationary and angular estimates). At multiplicity $2r + j$, the stationary law and the fixed-endpoint law are

$$\mathcal{L}_j \sim -\text{const}\, E^{1-1/(2r)}, \qquad \mathcal{C}_j \sim -\text{const}\, E^{1/2}.$$

Thus any fixed exponent $\gamma > 1/2$ works in the chord target at every finite multiplicity. For a stationary loss KE^{γ}, any fixed $\gamma < 1$ eventually becomes too weak as r grows. This comparison does not order the full hypotheses: chord estimates concern whole intervals, whereas Theorem 2.5 imposes a condition only on the stationary sets.

Proposition 7.4 (Sharpness of the energy scale and location)**.** *The exponent* $1/2$ *with an arbitrary bounded coefficient, or a bounded loss paid by endpoint energy, does not exclude collisions in general.*

Proof. The backward-heat field $H(T - s, x) = x^2 + 2s$ has a double collision, stationary predecessor $x = 0$, and zero-free rightward intervals for every $s > 0$. With $y = \ell \geq 0$,

$$e = E(T - s, 0) = 4s^2, \qquad \mathcal{C}_0 = 2s(2s + \ell^2 - 2\ell) \geq -2s = -\sqrt{e}.$$

For endpoint energy put $q = H_x(T - s, \ell)/H(T - s, \ell) \geq 0$ and $b = H(T - s, 0)/H(T - s, \ell) \in (0, 1]$. Then

$$\frac{\mathcal{C}_0}{E(T - s, \ell)} = b\frac{1 - q}{1 + q^2} \geq -\frac{1}{4}, \qquad 1 - q + \frac{1 + q^2}{4} = \frac{(q - 2)^2 + 1}{4} > 0.$$

For $q \leq 1$ the quotient is nonnegative; for $q > 1$, multiplication by $b \leq 1$ improves the lower bound. The still weaker payment by the sum of starting and endpoint energies also permits this collision. □

7.2 Transfer from a complete family of cutoff estimates

The chord criterion permits quantitative losses that need not be linear in E. For the transfer theorem in this subsection, choose the cutoff only after proving a single prescribed family of estimates for every integer cutoff. The endpoint-only criterion below has weaker cutoff requirements.

Proposition 7.5 (All-cutoff transfer)**.** *Let Q be a stationary observation with $e = E(t, x)$, or a chord observation with $e = E(t, x)$ at its starting point. Suppose*

$$Q \geq -Ae\exp(aN^{\beta}) - B\exp(-cN^2) \quad \text{for every integer } N \geq N_0, \tag{7.8}$$

where $A, a, B \geq 0$, $c > 0$, and $N_0 \geq 0$ is an integer. Assume fixed local upper bounds A_*, a_*, B_*, N_*, with $N_* \geq 1$, and a positive lower bound c_0 for c. For chords all these bounds are uniform over the admissible intervals of Theorem 7.2. The coefficient rules are prescribed before selecting the cutoff and are not fitted to an observed energy floor.

If $0 < \beta < 2$, set $M = \max(1, B_*)$, $L(e) = 1 + \max(0, \log(M/e))$, and $D = N_* + 1 + c_0^{-1/2}$. For $e > 0$,

$$Q \geq -(A_* + 1)e\exp\{a_* D^{\beta} L(e)^{\beta/2}\}. \tag{7.9}$$

This excludes finite collisions for the stationary target with its predecessor coverage, and for the chord target with its full interval coverage.

If $\beta = 2$ and $a_* < c_0$, put

$$\kappa = a_* + c_0, \qquad p = \frac{c_0}{\kappa} > \frac{1}{2}, \qquad D_2 = N_* + 1.$$

For $0 < e \le M$, with $L_2 = \log(M/e)$, one has

$$Q \ge -M^{1-p}e^p \left[A_* \exp\left\{ a_* D_2^2 + 2a_* D_2 \sqrt{L_2/\kappa} \right\} + 1 \right]. \tag{7.10}$$

This loss is $o(\sqrt{e})$ and therefore excludes collisions for the chord target. Equivalently, every $1/2 < \gamma < p$ yields a bound $Q \ge -K_\gamma e^\gamma$ for sufficiently small e.

Proof. For the first claim choose

$$N = \left\lceil \max\left(N_*, \sqrt{\max(0, \log(M/e))/c_0} \right) \right\rceil.$$

Then $B\exp(-cN^2) \le e$ and $N \le D\sqrt{L(e)}$, proving (7.9). Its relative loss is $\exp(o(\log(1/e)))$. Along (7.6) it is $s^{-o(1)}$, which cannot absorb the stationary pole r/s. Its absolute loss is also $o(\sqrt{e})$, so Theorem 7.2 applies.

For the Gaussian claim choose $N = \lceil \max(N_*, \sqrt{L_2/\kappa}) \rceil$. The inequalities $\sqrt{L_2/\kappa} \le N \le D_2 + \sqrt{L_2/\kappa}$ give (7.10) directly. The factor $\exp(O(\sqrt{\log(1/e)}))$ pays integer rounding and the cutoff threshold. It is dominated by $e^{-(p-\gamma)}$ for any fixed $\gamma < p$ as $e \downarrow 0$. At $e = 0$, letting $N \to \infty$ in (7.8) gives $Q \ge 0$. □

The strict local rate gap $a_* < c_0$ cannot be replaced here by a nonuniform pointwise gap that closes at the event. Nor may the rounding factor in (7.10) be dropped when claiming an exact $O(e^p)$ loss. At equality of the rates the preceding double-collision control satisfies, for every real N,

$$e\exp(N^2) + \exp(-N^2) \ge 2\sqrt{e} = 4s, \qquad \mathcal{L}_0 = -4s, \qquad \mathcal{C}_0 \ge -2s.$$

Thus both observations obey (7.8) with $\beta = 2$ and $A = B = a = c = 1$ despite the collision. These transfer statements are conditional analytic implications, not consequences of checking finitely many cutoffs. A separately truncated observation requires its own complete error estimate; a proved error $Q_0\sqrt{e}\exp(-c_1N^2)$ can be paid by $e/2 + (Q_0^2/2)\exp(-2c_1N^2)$.

7.3 Exponential averages and complete-start cutoff transfer

The endpoint derivative can be removed by a positive average without losing the collision obstruction. For $W = h_j$, $j = 0, 1$, define

$$\mathcal{A}_{j,\delta}(t,x) = W(t,x)^2 - e^{-\delta}W(t,x)W(t,x+\delta), \qquad \delta > 0. \tag{7.11}$$

The fundamental theorem of calculus gives, without stationarity,

$$\mathcal{A}_{j,\delta}(t,x) = \int_0^\delta e^{-z}\mathcal{C}_j(t;x,x+z)\,dz. \tag{7.12}$$

Indeed, $\partial_z(e^{-z}W(t,x+z)) = e^{-z}(W_x - W)(t,x+z)$. The normalization by $1 - e^{-\delta}$ is a positive probability average, but its density does not vanish at the endpoints; no rapid Fourier-decay claim for a compactly supported smooth density is used.

Proposition 7.6 (Fixed-length averaged collision law)**.** *At a multiplicity $2r + j$ event (T, a), $r \ge 1$, let $\chi(s)$ be the full stationary predecessor and set $a_0 = \partial_x^{2r}W(T,a)/(2r)! \ne 0$. For every sufficiently small fixed $\delta > 0$, the whole interval $[\chi(s), \chi(s) + \delta]$ is W-zero-free for sufficiently small $s > 0$, and*

$$\frac{\mathcal{A}_{j,\delta}(T-s,\chi(s))}{\sqrt{E(T-s,\chi(s))}} \longrightarrow -d_{j,\delta} < 0, \tag{7.13}$$

$$d_{j,\delta} = e^{-\delta}\operatorname{sgn}(a_0)W(T,a+\delta) = e^{-\delta}|a_0|\delta^{2r}(1+O(\delta)).$$

The energy is $E = H^2 + H_x^2$ on both branches.

Proof. The positive heat polynomial in the proof of Theorem 7.1 proves the whole-interval assertion, since $\chi(s) - a = O(s)$. Equation (7.6) gives $W/\sqrt{E} \to \operatorname{sgn}(a_0)$ on the predecessor. Moreover $W^2/\sqrt{E} \le \sqrt{E} \to 0$. Continuity at the moving endpoint, whose limit is $a + \delta$, proves the limit; the Taylor expansion $W(T, a + \delta) = a_0\delta^{2r} + O(\delta^{2r+1})$ proves its sign. □

For theta, let Φ_N be the first N summands of (2.1), and put

$$W_N(t, y) = \partial_y^j \int_0^\infty e^{tu^2} \Phi_N(u) \cos(yu)\, du.$$

Truncate only the endpoint:

$$\mathcal{A}_{j,\delta,N} = W(t, x)^2 - e^{-\delta} W(t, x) W_N(t, x + \delta). \tag{7.14}$$

Both the starting W and stationarity $h_{j+1}(t, x) = 0$ remain complete. For $t \le T_*$ and real y, the source estimate gives

$$|W(t, y) - W_N(t, y)| \le B_j(T_*) e^{-\pi(N+1)^2/4}, \tag{7.15}$$

$$B_j(T_*) = 2\pi^2 \left(\sum_{n \ge 1} n^4 e^{-\pi n^2/4} \right) \int_0^\infty u^j e^{T_* u^2 + 9u - (\pi/2)e^{4u}}\, du < \infty.$$

To check this bound, dominate the nth source summand by $2\pi^2 n^4 e^{9u - \pi n^2 e^{4u}}$ and use $n^2 e^{4u} \ge (n^2 + e^{4u})/2$. Splitting the remaining Gaussian sum into two equal exponent shares extracts $e^{-\pi(N+1)^2/4}$. Consequently

$$|\mathcal{A}_{j,\delta} - \mathcal{A}_{j,\delta,N}| \le e^{-\delta} B_j(T_*) e^{-\pi(N+1)^2/4} \sqrt{E(t, x)}. \tag{7.16}$$

Preserving the complete starting factor is what produces $\sqrt{E}$; a separately truncated starting factor does not inherit this estimate.

Theorem 7.7 (Fixed-length mixed-error exclusion)**.** *For the complete theta field, fix prescribed positive lengths $\delta_k \to 0$. Near each covered candidate event, for each branch $j = 0, 1$ and every sufficiently large fixed k, suppose there is a prescribed unbounded cutoff sequence $N_{k,\ell} \to \infty$ with estimates*

$$\mathcal{A}_{j,\delta_k,N_{k,\ell}}(t, x) \ge -K_{j,k,\ell}(t, x) E(t, x) - \varepsilon_{j,k,\ell} \sqrt{E(t, x)}. \tag{7.17}$$

Here $K_{j,k,\ell} \ge 0$ has a finite local upper bound through the event for each fixed pair (k, ℓ), and the nonnegative local error bounds satisfy $\varepsilon_{j,k,\ell} \to 0$ as $\ell \to \infty$ for each fixed j, k. The estimates hold on declared full-stationary domains containing all sufficiently close predecessors and their entire W-zero-free intervals of length δ_k. Each fixed-index time threshold may differ; the error bounds must be uniform over these approaching starts, and the coefficient rules must be prescribed independently of an observed positive energy floor. Then no covered finite multiple real zero occurs.

Proof. At a putative event choose its parity branch, and first fix k so that Proposition 7.6 gives $d_{j,\delta_k} > 0$ and the interval lies in the covered neighborhood. Next choose one fixed ℓ with

$$\varepsilon_{j,k,\ell} + e^{-\delta_k} B_j(T_*) e^{-\pi(N_{k,\ell}+1)^2/4} < d_{j,\delta_k}/2.$$

Combine (7.17) and (7.16), divide by $\sqrt{E}$, and let $s \downarrow 0$ along the full predecessor with both indices fixed. Local boundedness makes $K_{j,k,\ell}\sqrt{E} \to 0$. The lower bound contradicts (7.13). □

No growth restriction on $K_{j,k,\ell}$ as the indices increase is needed, and an unbounded certified cutoff subsequence suffices. The proof uses neither a simultaneous cutoff/time limit nor one constant uniform in spectral height or as the event time tends to zero. Every positive-time finite-height event must nevertheless be covered. For the complete theta field, coverage of $0 < T \le 1/2$ has the consequence stated in Remark 2.2.

If only one shrinking length δ_N is available per cutoff, the same argument works when the total mixed-error coefficient obeys $\varepsilon_N = o(\delta_N^m)$ for every fixed positive integer m. Indeed, choose a single N with error smaller than the collision signal $d_{j,\delta_N} \asymp \delta_N^{2r}$ before taking the time limit. A Gaussian total mixed error permits $\delta_N = N^{-q}$ for every fixed $q > 0$; an algebraic total error suffices for $\delta_N = 1/\log(N+2)$. The independent-cutoff version requires only convergence at each fixed length, but asks for a broader certificate domain. These refinements are developed in [18, 19, 27]. The theta core estimate (7.17) remains unproved; the tail bound alone supplies no favorable sign.

7.4 The inner profile and quarter-turn concentration

Lemma 7.8 (The real scaled heat profiles)**.** *Use the notation of Theorem 7.1. Uniformly on each fixed compact interval of ξ,*

$$\begin{aligned} W(T-s,\chi(s)+s\xi) &= ks^r\{1+O(s)\},\\ W_x(T-s,\chi(s)+s\xi) &= ks^r\{r\xi+O(s)\},\\ W_{xx}(T-s,\chi(s)+s\xi) &= ks^{r-1}\{r+O(s)\}. \end{aligned} \tag{7.18}$$

On each fixed interval $0 \le y \le c$, $c > 0$, one also has

$$W(T-s,\chi(s)+\sqrt{s}\,y) = ks^r\{F_r(y)+O(\sqrt{s})\}, \tag{7.19}$$

$$W_x(T-s,\chi(s)+\sqrt{s}\,y) = ks^{r-1/2}\{F_r'(y)+O(\sqrt{s})\}, \tag{7.20}$$

$$F_r(y) = \sum_{\ell=0}^{r} \frac{r!}{(r-\ell)!(2\ell)!} y^{2\ell}. \tag{7.21}$$

In particular, W has the sign of k on each such whole interval for small $s > 0$; $F_r \ge 1$ and $F_r'(c) > 0$.

Proof. In (2.9) put $x - a = s(\eta(s)+\xi)$, where $\eta(s) = (\chi(s)-a)/s$ is analytic. The leading coefficients of W, W_x, W_{xx} after removing respectively s^r, s^r, s^{r-1} are $k, kr\xi, kr$; for W_x, the constant term cancels because $\eta(0) = -\partial_x^{2r+1}W(T,a)/(r\partial_x^{2r}W(T,a))$. The resulting analytic expansions and their fixed-order derivatives are uniform on compact ξ intervals. For the parabolic scale put $s = \eta^2$ instead. The coefficient of η^{2r} in $W(T-\eta^2,\chi(\eta^2)+\eta y)$ is $kF_r(y)$; the displacement $\chi(\eta^2)-a = O(\eta^2)$ does not affect it. Uniform analytic remainders and their first y derivatives prove the two expansions. Positivity of the displayed polynomial proves the sign assertions. □

Equations (6.4) and (7.18) give the finer angular core, uniformly on compact ξ intervals:

$$\begin{aligned} \alpha_j(T-s,\chi(s)+s\xi) - \alpha_j(T-s,\chi(s)) &= \arctan(r\xi)+O(s),\\ s\,\partial_x\alpha_j(T-s,\chi(s)+s\xi) &= \frac{r}{1+r^2\xi^2}+O(s). \end{aligned} \tag{7.22}$$

These compact-window limits assert no weak limit on a fixed spatial neighborhood and require no unproved uniform tail estimate. For an odd collision the same Taylor counting gives $H(T-s,\chi(s)+s\xi) = O(s^{r+1})$, so

$$\frac{E_1^{\mathrm{jet}}}{E}(T-s,\chi(s)+s\xi) \longrightarrow 1+r^2\xi^2. \tag{7.23}$$

Thus agreement of the two energies at the stationary center cannot be extended across a nonzero inner window without retaining this factor.

Theorem 7.9 (Positive quarter-turn concentration)**.** *For each fixed $c > 0$, the collision and predecessor above satisfy*

$$\alpha_j(T-s, \chi(s) + c\sqrt{s}) - \alpha_j(T-s, \chi(s)) \longrightarrow \frac{\pi}{2}. \tag{7.24}$$

The defining pair is regular on the whole interval, including on branch $j = 1$. The chord limits at the two shrinking scales are

$$\begin{aligned} \frac{\mathcal{C}_j(T-s; \chi(s), \chi(s) + Cs/r)}{E(T-s, \chi(s))} &\longrightarrow 1 - C \quad (C > 0), \\ \frac{\mathcal{C}_j(T-s; \chi(s), \chi(s) + c\sqrt{s})}{E(T-s, \chi(s))} &= -\frac{F_r'(c)}{\sqrt{s}} + O(1). \end{aligned} \tag{7.25}$$

Proof. Zero-freeness follows from Lemma 7.8. At the parabolic endpoint,

$$W_x/W = s^{-1/2}(F_r'(c)/F_r(c))(1 + O(\sqrt{s})) \to +\infty.$$

Equation (7.1) proves the angle limit with either sign of k. Substitution of the two jet expansions into (7.2), using (7.6), proves both chord limits. □

The finite inner limit alone can be absorbed by a growing relative energy allowance. A bound uniform on all short intervals must also hold on the parabolic windows and at the fixed nearby endpoint, where the stronger obstructions apply. This interval quantifier explains why the cutoff conclusions above are stronger than a test on one fixed inner window; see [10].

7.5 A supporting one-sided continuity criterion

Definition 7.10 (One-sided modulus from stationary starting points)**.** A field has a local one-sided turning modulus on a declared domain if, near each covered candidate event, there are a fixed $\ell_0 > 0$ and a prescribed nonnegative function $\omega(\ell) \to 0$ as $\ell \downarrow 0$, uniform in time and starting point, such that for $j = 0, 1$

$$\alpha_j(t, x + \ell) - \alpha_j(t, x) \le \omega(\ell) \tag{7.26}$$

whenever $0 < \ell \le \ell_0$, $h_{j+1}(t, x) = 0$, $h_j(t, x) \neq 0$, the segment lies in the covered neighborhood, and $(h_j, h_{j+1}) \neq (0, 0)$ throughout it. The increment uses its continuous lift along the segment.

Theorem 7.11 (Oriented-continuity exclusion)**.** *Let H be a real analytic backward-heat field with nonzero spatial slices. If Definition 7.10 holds with coverage of the sufficiently close full predecessors and their short rightward intervals, then no covered finite multiple real zero occurs. For the complete theta field, coverage of $0 < T \le 1/2$ implies RH by Remark 2.2.*

Proof. At a candidate event take $\ell = c\sqrt{s}$ on its parity branch. Lemma 7.8 supplies the required regularity on the whole interval. The upper bound tends to zero, contradicting Theorem 7.9. □

On every regular test interval the observed increment is exactly

$$\int_x^{x+\ell} \frac{h_j h_{j+2} - h_{j+1}^2}{h_j^2 + h_{j+1}^2}(t, y)\, dy. \tag{7.27}$$

A uniform upper bound on the angular derivative throughout these intervals implies a linear modulus. More generally a uniform strict gap below $\pi/2$ on all sufficiently short regular intervals already contradicts (7.24). Both formulations retain the orientation and allow negative turning; neither follows from a curvature-magnitude bound or from continuity on individual regular slices. No global regularity of the second branch is assumed.

The resulting hierarchy identifies concrete conditional source targets. The averaged criterion in Theorem 7.7 retains the sharp starting-energy scale while relaxing the endpoint observations and cutoff growth required of a certificate. The exact polynomial controls in [9, 10] test the local identities and their scope; they establish no sign bound for the theta source. The remaining task is a complete-theta estimate with the required signs, starting-point energy, and uniform interval coverage.

8 The complete theta source on a curved elliptic family

The preceding geometric realizations retain the physical jets, but a local realization alone does not distinguish the theta source from a collision control. We now place the complete source in a family carrying its actual integer lattice. There are three separate structures: the holomorphic theta line bundle, the physical heat-weighted transform, and a nonflat hyperkähler metric on the elliptic family. Their relations below are exact. They do not yet give the signed estimate at the physical energy scale. Our source normalization is that of [1].

8.1 The complete primitive and its boundary contribution

For $u \geq 0$ set

$$\Phi(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \qquad H(t,x) = \int_0^\infty e^{tu^2} \Phi(u) \cos(xu)\, du. \tag{8.1}$$

Every fixed source derivative decays faster than any Gaussian as $u \to +\infty$. The sums and integrals below therefore converge locally uniformly with all stated derivatives. In particular, H is entire in $(t,x) \in \mathbb{C}^2$ and satisfies $H_t = -H_{xx}$.

Write

$$\vartheta(z,\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}, \qquad \Im\tau > 0, \quad P(u) = e^u \vartheta(0, ie^{4u}), \quad \psi(u) = e^u \sum_{n\geq 1} e^{-\pi n^2 e^{4u}}. \tag{8.2}$$

Thus $P = e^u + 2\psi$ on the half-line. The Gaussian Poisson identity $\vartheta(0, i/y) = \sqrt{y}\,\vartheta(0, iy)$ implies $P(-u) = P(u)$. The function P grows like e^u at positive infinity; it is ψ, not P, that supplies a convergent positive-time primitive transform.

Proposition 8.1 (Complete differential reconstruction)**.** *The complete source and primitive satisfy*

$$\Phi = \frac{P'' - P}{16} = \frac{\psi'' - \psi}{8}, \qquad \psi^{(2k+1)}(0) = -\frac{1}{2} \quad (k \geq 0). \tag{8.3}$$

For

$$J(t,x) = \int_0^\infty e^{tu^2} \psi(u) \cos(xu)\, du \tag{8.4}$$

one has the exact identity

$$\boxed{8H = \frac{1}{2} + (2t - x^2 - 1)J + 4txJ_x - 4t^2 J_{xx}.} \tag{8.5}$$

All the theta indices occur in these identities.

Proof. Termwise differentiation of P gives the first identity in (8.3); $(e^u)'' - e^u = 0$ gives the second expression. Evenness of $P = e^u + 2\psi$ gives every displayed endpoint derivative. For $F(u) = e^{tu^2}\cos(xu)$, two integrations by parts yield

$$\int_0^\infty \psi'' F\,du = -\psi'(0)F(0) + \psi(0)F'(0) + \int_0^\infty \psi F''\,du.$$

The upper boundary terms vanish, $F(0) = 1$, and $F'(0) = 0$. Directly,

$$F'' - F = e^{tu^2}\left[(2t + 4t^2u^2 - x^2 - 1)\cos(xu) - 4txu\sin(xu)\right].$$

Insert $\psi'(0) = -1/2$ and differentiate (8.4). This gives (8.5), including its boundary constant. □

Proposition 8.2 (Positive primitive; signed reconstruction)**.** *For $0 \le t \le 1/2$ and every real x, $J(t,x) > 0$. Uniformly on this heat interval,*

$$J(t,x) = \frac{1}{2x^2} - \frac{1+6t}{2x^4} + O(x^{-6}) \qquad (|x| \to \infty). \tag{8.6}$$

At time zero the exact reconstruction is

$$J(0,x) = \frac{1}{2(1+x^2)} - \frac{8H(0,x)}{1+x^2}. \tag{8.7}$$

Proof. Let $q_n = \pi n^2 e^{4u}$ and $f_n(u) = \exp(tu^2 + u - q_n)$, so that $f = e^{tu^2}\psi = \sum_{n\ge1} f_n$. For $u \ge 0$ and $0 \le t \le 1/2$, $2tu + 1 \le u + 1 \le q_n$. Hence

$$\frac{f_n''}{f_n} = (2tu + 1 - 4q_n)^2 + 2t - 16q_n \ge 9q_n^2 - 16q_n > 0.$$

Thus $f'' > 0$. Moreover $f'(0) = -1/2$ and $f'(\infty) = 0$. For $x \ne 0$, integration by parts gives

$$J(t,x) = \frac{1}{x^2}\int_0^\infty f''(u)(1 - \cos(xu))\,du > 0. \tag{8.8}$$

At $x = 0$ the defining integral is positive. Repeated integration by parts uses $f'(0) = -1/2$ and $f'''(0) = -(1+6t)/2$ to give (8.6). The sixth derivative is uniformly integrable on the stated compact heat interval, which bounds its remainder. Finally (8.7) is (8.5) with $t = 0$. □

Remark 8.3 (What this positivity does not prove)*.* The physical field is reconstructed from a signed differential expression in a strictly positive channel. At $t = 0$ it is the difference between that channel and an explicit rational profile. The primitive-compatible collision controls of [3], have the same boundary constant, strict primitive positivity, and every algebraic asymptotic coefficient of J, while possessing a positive-time double zero. Their source is not the integer theta series. Accordingly, the next construction retains the theta line bundle and its section, rather than using primitive positivity alone.

8.2 The theta line bundle, its connection, and the fourth jet

Use the marked elliptic family

$$\mathcal{X} = (\mathbb{H} \times \mathbb{C})/\mathbb{Z}^2, \qquad (m,n) : (\tau, z) \longmapsto (\tau, z + m + n\tau),$$

where $\mathbb{H}$ is the upper half-plane. Write $\tau = p + iq$, $z = r + \tau s$, so that $q > 0$ and r, s are real fiber coordinates modulo one. The factors

$$j_{m,n}(\tau, z) = e^{-\pi i n^2 \tau - 2\pi i n z} \tag{8.9}$$

define a holomorphic line bundle $L \to \mathcal{X}$, and reindexing the series gives

$$\vartheta(z + m + n\tau, \tau) = j_{m,n}(\tau, z)\vartheta(z, \tau), \qquad 4\pi i\, \partial_\tau \vartheta = \partial_z^2 \vartheta. \tag{8.10}$$

The second equality follows term by term. These standard theta-section and heat-connection structures are described in [6, Section 6]; the explicit normalizations needed here are derived below.

Ordinary derivatives are jets, not separate copies of the same holomorphic section. For every nonnegative integer k,

$$(\partial_z^k \vartheta)(z + m + n\tau, \tau) = j_{m,n}(\tau, z) \sum_{a=0}^{k} \binom{k}{a} (-2\pi i n)^{k-a} (\partial_z^a \vartheta)(z, \tau). \tag{8.11}$$

Thus a global description must keep the lower-jet terms or choose a connection. The following choice also identifies the actual curvature of the theta line bundle.

Proposition 8.4 (Hermitian theta connection)**.** *In the holomorphic frame on $\mathbb{H} \times \mathbb{C}$, the metric*

$$h(\tau, z) = \exp\left(-\frac{2\pi(\Im z)^2}{q}\right) \tag{8.12}$$

descends to L. Its Chern connection and curvature are

$$\nabla = d + A, \qquad A = \partial \log h = 2\pi i s\, dz - \pi i s^2\, d\tau, \tag{8.13}$$

$$F_\nabla = \frac{\pi}{q} \eta \wedge \overline{\eta} = -2\pi i\, dr \wedge ds, \qquad \eta = dz - s\, d\tau = dr + \tau ds. \tag{8.14}$$

In particular, $\int_{E_\tau} iF_\nabla/(2\pi) = 1$. The vertical covariant operator in the chosen relative frame is

$$\mathcal{D} = \nabla_{\partial_z} = \partial_z + 2\pi i s. \tag{8.15}$$

Each $\mathcal{D}^k \vartheta$ transforms by the multiplier (8.9).

Proof. Put $v = \Im z$. Under a lattice translation v becomes $v + nq$, while $|j_{m,n}|^2 = e^{2\pi n^2 q + 4\pi n v}$. Consequently $h(\tau, z + m + n\tau)|j_{m,n}(\tau, z)|^2 = h(\tau, z)$. Using $\partial_z v = 1/(2i)$ and $\partial_\tau q = 1/(2i)$ gives (8.13). Differentiation yields

$$dA = 2\pi i\, ds \wedge (dz - s\, d\tau).$$

Since $ds = (\eta - \overline{\eta})/(2iq)$, this is $(\pi/q)\eta \wedge \overline{\eta}$. Finally $\eta \wedge \overline{\eta} = -2iq\, dr \wedge ds$. The fiber integral is one, with orientation $dr \wedge ds$. Under translation s becomes $s + n$ and $\partial_z j_{m,n} = -2\pi i n j_{m,n}$. The two extra terms cancel in $\mathcal{D}(j_{m,n} a)$ after using the translated s. Iterating proves the covariant transformation statement. □

Remark 8.5. The covariant derivatives here are smooth sections. They need not be holomorphic: $[\partial_{\bar{z}}, \mathcal{D}] = -\pi/q$. Also, $\mathcal{D}^k$ means repeated vertical differentiation in the translation-invariant frame ∂_z of the relative tangent bundle. It is not an unspecified full covariant Hessian on the total space. The Chern connection is distinct from the theta heat connection in (8.10).

Theorem 8.6 (Exact covariant theta-source formula)**.** *Let $y = e^{4u}$ and evaluate the theta section at $(\tau, z) = (iy, 0)$. Then*

$$\Phi(u) = y^{1/4}\left(y^2\partial_y^2 + \frac{3}{2}y\partial_y\right)\vartheta(0, iy) \tag{8.16}$$

$$= y^{1/4}\left(\frac{y^2}{16\pi^2}\partial_z^4 + \frac{3y}{8\pi}\partial_z^2\right)\vartheta(z, iy)\big|_{z=0} \tag{8.17}$$

$$= \boxed{\frac{y^{1/4}}{16\pi^2}(y^2\mathcal{D}^4\vartheta - 3\pi^2\vartheta)(0, iy).} \tag{8.18}$$

The final expression is evaluated in the stated holomorphic frame along the zero section of the marked family. It is a source identity, not a norm identity or a positivity statement.

Proof. For $D_y = y\partial_y$, one has $\partial_u = 4D_y$ and $y^{-1/4}D_y y^{1/4} = D_y + 1/4$. Thus

$$\frac{1}{16}y^{-1/4}(\partial_u^2 - 1)y^{1/4} = D_y^2 + \tfrac{1}{2}D_y = y^2\partial_y^2 + \tfrac{3}{2}y\partial_y,$$

which proves (8.16). The theta heat equation gives $\partial_y\vartheta(z, iy) = (4\pi)^{-1}\partial_z^2\vartheta(z, iy)$, proving (8.17). At fixed τ, let $a = 2\pi is$, so $\mathcal{D} = \partial_z + a$. Then $\partial_z a = \pi/q$ and $\partial_z^2 a = 0$. At $z = 0$, $a = 0$, and the operator product gives

$$(\mathcal{D}^2\vartheta)|_0 = (\partial_z^2\vartheta)|_0 + \frac{\pi}{q}\vartheta|_0,$$
$$(\mathcal{D}^4\vartheta)|_0 = (\partial_z^4\vartheta)|_0 + \frac{6\pi}{q}(\partial_z^2\vartheta)|_0 + \frac{3\pi^2}{q^2}\vartheta|_0.$$

Set $q = y$ and substitute the last equality into (8.17). The second-derivative terms cancel, leaving (8.18). □

The physical reconstruction is now an exact integral along the real path $\tau = iy$, $y \geq 1$, in this marked family:

$$H(t, x) = \frac{1}{4}\int_1^\infty e^{t(\log y)^2/16}y^{1/4}\left(y^2\partial_y^2 + \frac{3}{2}y\partial_y\right)\vartheta(0, iy)\cos\left(\frac{x}{4}\log y\right)\frac{dy}{y} \tag{8.19}$$

$$= \frac{1}{64\pi^2}\int_1^\infty e^{t(\log y)^2/16}y^{1/4}(y^2\mathcal{D}^4\vartheta - 3\pi^2\vartheta)(0, iy)\cos\left(\frac{x}{4}\log y\right)\frac{dy}{y}. \tag{8.20}$$

Both integrals retain all theta indices. The physical heat multiplier $e^{t(\log y)^2/16}$ is not the parameter evolution in the Jacobi heat equation. Inversion $y \mapsto 1/y$ preserves this squared-log multiplier; that reflection symmetry must not be discarded. Nevertheless, the oscillatory integral (8.20) does not become a positive Hermitian norm merely because L has positive fiber degree.

8.3 A nonflat hyperkähler metric with the correct elliptic fibers

The same marked family has an explicit metric which incorporates its varying elliptic modulus. This is a local semi-flat construction in the sense of [7, Section 2]; we give a direct verification so that neither its curvature nor its global scope is implicit.

Proposition 8.7 (Semi-flat geometry on the marked cover)**.** *On* $\mathbb{H}\times(\mathbb{R}/\mathbb{Z})^2$ *the metric*

$$g = q(dp^2+dq^2)+q^{-1}(dr+p\,ds)^2+q\,ds^2 = q|d\tau|^2+q^{-1}|\eta|^2 \tag{8.21}$$

is hyperkähler and nonflat. Its elliptic fibers have modulus τ *and area one. A parallel hyperkähler triple is*

$$\omega_I = q\,dp\wedge dq+dr\wedge ds, \qquad \omega_J+i\omega_K = d\tau\wedge dz. \tag{8.22}$$

For the orientation $\omega_I^2/2>0$*, the three forms are self-dual. The horizontal sectional curvature is* $1/(2q^3)$*.*

Proof. The fiber metric is $q^{-1}|dr+\tau ds|^2$, with area form $dr\wedge ds$. Introduce the orthonormal coframe

$$e^1=\sqrt{q}\,dp, \quad e^2=\sqrt{q}\,dq, \quad e^3=q^{-1/2}(dr+p\,ds), \quad e^4=\sqrt{q}\,ds.$$

Writing $e^{ab}=e^a\wedge e^b$, direct expansion gives

$$\omega_I = e^{12}+e^{34}, \qquad \omega_J = e^{13}-e^{24}, \qquad \omega_K = e^{14}+e^{23}, \qquad \omega_a\wedge\omega_b = 2\delta_{ab}\,\mathrm{vol}_g\,. \tag{8.23}$$

The first form is closed. Moreover $d\tau\wedge dz = d\tau\wedge(dr+\tau ds)$, which is closed, so the other two forms are closed as well.

For a direct Kähler verification, (τ,z) are holomorphic coordinates on a local lift, and

$$\omega_I = \frac{i}{2}(q\,d\tau\wedge d\bar\tau+q^{-1}\eta\wedge\bar\eta).$$

The Hermitian coefficient matrix of the metric is

$$G=\begin{pmatrix} q+s^2/q & -s/q \\ -s/q & 1/q\end{pmatrix}, \qquad \det G=1. \tag{8.24}$$

Thus the Kähler Ricci form $-i\partial\bar\partial\log\det G$ vanishes. The induced Chern connection on the canonical bundle has zero connection form in the frame $d\tau\wedge dz$; since the metric is Kähler, this frame is Levi-Civita parallel. Its real and imaginary parts, together with the parallel Kähler form, give the parallel triple (8.22). The algebra (8.23) fixes its normalization and orientation.

Finally, the surfaces with r,s constant are totally geodesic: in these real coordinates the metric is block diagonal, and its base block is independent of r,s. Their induced metric is $q(dp^2+dq^2)$. Its Gaussian curvature is

$$K_{\text{base}} = -\frac{1}{2q}(\partial_p^2+\partial_q^2)\log q = \frac{1}{2q^3}>0. \tag{8.25}$$

Total geodesy identifies this with the ambient sectional curvature of the horizontal plane, proving nonflatness. □

Remark 8.8 (The precise global scope)*.* The formulas descend under the fiber lattice and are defined over the whole marked upper half-plane. They do not give a metric on the full modular quotient. Under

$$\tau'=\frac{a\tau+b}{c\tau+d}, \qquad z'=\frac{z}{c\tau+d}, \qquad ad-bc=1,$$

one has

$$d\tau'\wedge dz' = (c\tau+d)^{-3}d\tau\wedge dz, \qquad q'|d\tau'|^2 = |c\tau+d|^{-6}q|d\tau|^2. \tag{8.26}$$

The base metric therefore fails modular invariance as written. There is also incompleteness: a path with p,r,s constant and $q\downarrow 0$ has finite length $\int_0^{q_0}\sqrt{q}\,dq$. No complete K3 metric or extension through singular fibers is asserted. The global construction in [7] introduces local models and corrections around singular elliptic fibers; that separate existence theorem does not identify the physical transform with a K3 invariant. Any modular descent here must supply the missing base-period and line-bundle data and then control the resulting boundary terms.

8.4 An actual abelian anti-self-dual theta connection

The connection in Proposition 8.4 carries the arithmetic section, but its curvature is not anti-self-dual for (8.21). This distinction can be resolved explicitly on the marked noncompact family, without changing the vertical source formula.

Proposition 8.9 (A Chern connection with anti-self-dual curvature)**.** *Let $\omega_b = q\,dp\wedge dq$ and $\omega_f = dr\wedge ds$. Equip the same holomorphic theta line bundle with the Hermitian metric*

$$\widehat{h} = \exp\!\left(\frac{2\pi}{3}q^3\right) h. \tag{8.27}$$

Its Chern connection has connection form and curvature

$$\widehat{A} = A - \pi i q^2\, d\tau, \tag{8.28}$$

$$F_{\widehat{\nabla}} = 2\pi i(\omega_b - \omega_f), \qquad *F_{\widehat{\nabla}} = -F_{\widehat{\nabla}}. \tag{8.29}$$

It is an abelian Yang–Mills connection, and its vertical covariant operator remains $\mathcal{D}$. Thus (8.18) is unchanged.

Proof. The extra factor in (8.27) depends only on the base, so it preserves the lattice descent and the holomorphic bundle. Since $\partial_\tau q = 1/(2i)$,

$$\partial\!\left(\frac{2\pi}{3}q^3\right) = -\pi i q^2\, d\tau, \qquad d(-\pi i q^2\, d\tau) = 2\pi i q\, dp\wedge dq.$$

Combine this with (8.14). In the orthonormal coframe, $\omega_b = e^{12}$ and $\omega_f = e^{34}$, so $*\omega_b = \omega_f$ and $*\omega_f = \omega_b$. This proves anti-self-duality. The curvature is closed, and hence $d * F_{\widehat{\nabla}} = 0$; for a line bundle this is the Yang–Mills equation. The extra connection form vanishes on ∂_z, proving the last assertion. □

Remark 8.10 (Scope of the ASD construction). This is a nonflat geometric setting with the actual theta section and an anti-self-dual line-bundle connection. The unmodified Chern curvature has self-dual part $F^+_\nabla = -\pi i\omega_I$. The base rescaling removes that part while preserving the vertical source expression. It provides neither compactness nor a finite-energy instanton: the real curvature $F_{\widehat{\nabla}}/i$ has squared norm $8\pi^2$, while $\mathrm{vol}_g = q\,dp\,dq\,dr\,ds$ has infinite integral on this cover. The displayed metric's rapid cusp growth is not an invariant obstruction; the classification below gives a representative normalized on the physical path. Neither positive degree nor the Yang–Mills equation supplies the required signed physical estimate.

8.5 Physical restriction and complete signed source identities

The ASD construction does not determine the norm of the distinguished arithmetic section, and its physical restriction is flat. The useful data that survive are the transverse jet identities and the complete signed transform, including its endpoints [11, 12].

Proposition 8.11 (Metric freedom and transverse jets)**.** *Among base-only changes $h_b = e^{b(p,q)}h$, the ASD metrics are exactly*

$$b = \frac{2\pi}{3}q^3 + 2\Re f(\tau), \qquad f \text{ holomorphic on } \mathbb{H}. \tag{8.30}$$

They have the same curvature and vertical operator $\mathcal{D}$. In particular, $f = -\pi i\tau^3/3$ gives

$$\widetilde{h} = e^{2\pi p^2 q} h, \qquad \widetilde{A} = A - \pi i(q^2+\tau^2)d\tau, \qquad \gamma^*\widetilde{h} = 1, \quad \gamma^*\widetilde{A} = 0, \tag{8.31}$$

where $\gamma(u) = (ie^{4u}, 0)$. The natural external product connection on two physical paths is also flat.

In a marked holomorphic chart set $X = 4q\partial_q|_{p,z,\bar{z}}$ and $J_k(u) = (\mathcal{D}^k\vartheta)(0, iq)$, $q = e^{4u}$. Compute the commutators before restriction to the zero section. Then

$$[\nabla_X^{(b)}, \mathcal{D}^k] = -4k\pi i s\mathcal{D}^{k-1} - \frac{k(k-1)\pi}{q}\mathcal{D}^{k-2}, \tag{8.32}$$

$$J_k' = \frac{q}{\pi}J_{k+2} - (2k+1)J_k - \frac{k(k-1)\pi}{q}J_{k-2}. \tag{8.33}$$

Terms with zero prefactor are omitted. The extension X is chart-local; it is not asserted to be a global horizontal field under the τ-dependent lattice translations.

Proof. Direct differentiation gives

$$F_b = -2\pi i\, dr \wedge ds + \frac{i}{2}(b_{pp} + b_{qq})\, dp \wedge dq.$$

Thus ASD is equivalent to $\Delta b = 4\pi q$. The difference from $2\pi q^3/3$ is harmonic, and the simply connected upper half-plane supplies its harmonic conjugate. The stated choice of f gives (8.31). For comparison, $\gamma^*\widehat{A} = d(\pi e^{12u}/3)$; on the external product the connection is the sum of these two exact one-forms, with zero curvature.

Write $a = 2\pi i s$ and $c = \pi/q$. One has $[\nabla_X^{(b)}, \mathcal{D}] = -2a$, $[\mathcal{D}, a] = c$, and $[\mathcal{D}, c] = 0$; the base-only connection term commutes with $\mathcal{D}$. Iteration gives (8.32). The recurrence follows by differentiating the generating identity

$$\sum_{k\geq 0} J_k(u)\frac{w^k}{k!} = e^{\pi w^2/(2q)}\vartheta(w, iq)$$

and using the Jacobi heat equation. The base scalar cancels upon returning to coefficients in the marked arithmetic frame. □

The rapid cusp growth of $\widehat{h}$ is therefore a property of that specified metric, not an obstruction determined by ASD curvature. Changing the metric while keeping ϑ fixed changes its norm; the equivalent holomorphic bundle isometry instead transforms both the section and its dual observation. Flat physical restriction does not remove the second term of (8.32): for $k \geq 2$ it survives at $s = 0$. It also does not make the oscillatory observation a Hermitian norm.

Proposition 8.12 (The complete signed bilocal observations)**.** *Put $c_j(x,u) = u^j\cos(xu + j\pi/2)$ and define the chord*

$$\mathcal{C}_j(t;x,y) = h_j(t,x)\{h_j(t,y) - h_{j+1}(t,y)\}, \qquad j = 0, 1.$$

For $x < y$, $h_{j+1}(t,x) = 0$, and h_j nonzero on the whole interval, its nonnegativity is equivalent to an angular increase of at most $\pi/4$. Define the symmetric source kernels

$$K_j(x;u,v) = \frac{(uv)^j}{4}\left[(u+v)^2\cos(x(u-v)) + (-1)^j(u-v)^2\cos(x(u+v))\right], \tag{8.34}$$

$$M(x;u,v) = \frac{1}{2}\left[(1+uv)\cos(x(u-v)) + (1-uv)\cos(x(u+v))\right], \tag{8.35}$$

$$T_j(x,y;u,v)=\frac{1}{2}[c_j(x,u)\{c_j(y,v)-c_{j+1}(y,v)\} + c_j(x,v)\{c_j(y,u)-c_{j+1}(y,u)\}]. \tag{8.36}$$

With $\mathcal{B}(u)=e^{8u}J_4(u)-3\pi^2J_0(u)$, all three observations have the same complete covariant insertion:

$$\begin{pmatrix}\mathcal{L}_j(t,x)\\ E(t,x)\\ \mathcal{C}_j(t;x,y)\end{pmatrix}=\frac{1}{256\pi^4}\int_0^\infty\int_0^\infty e^{t(u^2+v^2)+u+v}\mathcal{B}(u)\mathcal{B}(v)\begin{pmatrix}K_j(x;u,v)\\ M(x;u,v)\\ T_j(x,y;u,v)\end{pmatrix}du\,dv. \tag{8.37}$$

Proof. Write $h_j=\int_0^\infty e^{tu^2}\Phi(u)c_j(x,u)\,du$. Multiply the defining observations and symmetrize in u,v. The product-to-sum identities give K_j and M; the directed chord gives T_j. Insert (8.18) in both variables. Complete-source decay justifies every product and differentiation locally uniformly in the parameters. The angular statement follows from the continuous lift $\alpha_j(t,y)-\alpha_j(t,x)=\arctan(h_{j+1}(t,y)/h_j(t,y))$, because h_j has one sign throughout the interval. □

The primitive in (8.2) displays every lower-boundary contribution without splitting the cancelling theta terms. Define

$$I_j(t,x)=\int_0^\infty\psi(u)(\partial_u^2-1)\{e^{tu^2}c_j(x,u)\}\,du. \tag{8.38}$$

Two integrations by parts, using $\psi'(0)=-1/2$, give

$$\int_0^\infty(\psi''-\psi)f\,du=\tfrac{1}{2}f(0)+\psi(0)f'(0)+\int_0^\infty\psi(f''-f)\,du.$$

Here $f=e^{tu^2}c_j(x,u)$ is even, $f(0)=\delta_{j0}$, and the upper endpoints vanish by double exponential decay. Consequently $8h_j=I_j+\delta_{j0}/2$, and the full signed reconstructions are

$$64\mathcal{L}_0=I_1^2-I_0I_2-\tfrac{1}{2}I_2,\qquad 64\mathcal{L}_1=I_2^2-I_1I_3, \tag{8.39}$$

$$64E=\tfrac{1}{4}+I_0+I_0^2+I_1^2, \tag{8.40}$$

$$64\mathcal{C}_0(t;x,y)=I_0(t,x)\{I_0(t,y)-I_1(t,y)\}+\tfrac{1}{2}I_0(t,x)+\tfrac{1}{2}\{I_0(t,y)-I_1(t,y)\}+\tfrac{1}{4}, \tag{8.41}$$

$$64\mathcal{C}_1(t;x,y)=I_1(t,x)\{I_1(t,y)-I_2(t,y)\}. \tag{8.42}$$

Unspecified arguments in (8.39)–(8.40) are (t,x). The affine terms and constants record the lower edges and corner of the bilocal integration by parts; none are disposable. Every I_j retains the physical heat multiplier inside its differential operator. These identities specify complete quantities to estimate, but assert no sign. In a chord estimate the comparison energy is $E(t,x)$ at the starting point, also on branch $j=1$; neither endpoint energy nor the auxiliary norm $h_1^2+h_2^2$ may replace it.

9 Arithmetic discrimination: support, endpoints and reflection

The signed bridge requires information that distinguishes the actual theta section from positive weighted square sums. The following control rejects support-only moment certificates, even with the first source endpoint and primitive normalization imposed. Complete reflection, by contrast, forces the theta coefficients themselves. Neither statement proves the remaining signed estimate.

9.1 A validated collision within the square-support relaxation

Put $\lambda_n = \pi n^2$, $q = e^{4u}$, and

$$\phi_n(u) = e^u(2\lambda_n^2 q^2 - 3\lambda_n q)e^{-\lambda_n q}, \qquad H_n(t,x) = \int_0^\infty e^{tu^2}\phi_n(u)\cos(xu)\,du.$$

For fixed nonnegative weights a_n, write $\Phi_a = \sum_{n\ge1} a_n\phi_n$, $\psi_a = e^u\sum_{n\ge1} a_n e^{-\lambda_n q}$, and $H_a = \sum a_n H_n$. For finite support these definitions give entire physical transforms, $H_{a,t} = -H_{a,xx}$, and $8\Phi_a = \psi_a'' - \psi_a$. The source is positive on $u \ge 0$ whenever some weight is positive. Moreover, for every real polynomial V and every polynomial Q nonnegative on the square nodes,

$$\sum_{n\ge1} a_n e^{-\lambda_n q}Q(\lambda_n)V(\lambda_n)^2 \ge 0 \qquad (q>0). \tag{9.1}$$

This includes every finite moment matrix and adjacent-square localizing matrix, by taking $Q(\lambda) = (\lambda-\lambda_m)(\lambda-\lambda_{m+1})$.

Proposition 9.1 (Validated weighted-square control)**.** *There are fixed positive weights on $n = 1, 2, 3$, zero otherwise, and a fixed positive normalization such that*

$$\Phi_a'(0) = 0, \qquad \psi_a'(0) = \psi_a'''(0) = -\tfrac12, \qquad H_a(\tfrac14, x_*) = H_{a,x}(\tfrac14, x_*) = 0, \quad H_{a,xx}(\tfrac14, x_*) \ne 0,$$

where $44.1303 < x_* < 44.1304$. The third source derivative is nonzero, so this source is not even in u. The control satisfies (9.1) at every degree and every $q > 0$.

Proof. The construction and validated record are given in [14]. Define

$$P_1(a) = -8a^3 + 30a^2 - 15a, \quad d_n = P_1(\lambda_n), \quad G_n = e^{\lambda_n}H_n, \quad b_n = -d_1/d_n \quad (n = 2, 3).$$

Here $d_1 > 0 > d_2, d_3$, and $G_\theta = G_1 + \theta b_2 G_2 + (1-\theta)b_3 G_3$ has positive weights for $0 < \theta < 1$. Its source derivative at zero is exactly $d_1 + \theta b_2 d_2 + (1-\theta)b_3 d_3 = 0$. At $T = 1/4$, set $A = G_1 + b_3 G_3$, $B = b_2 G_2 - b_3 G_3$, and $D = A'B - AB'$. On $I = [44.1303, 44.1304]$, the validated enclosures include

$$A < 0 < B, \qquad D(44.1303) \in (5.0206, 5.0208)10^{-14},$$
$$D' \in (-1.178, -1.121)10^{-9}, \qquad D(44.1304) \in (-6.4752, -6.4750)10^{-14}.$$

Thus D has exactly one zero x_* in I. At that zero, $\theta_* = -A(x_*)/B(x_*) \in (0.87916, 0.87919)$ gives a double zero, with

$$\partial_x^2 G_{\theta_*}(T, x_*) \in (-5.95, -5.65)10^{-6}.$$

The raw weights are $(a_1, a_2, a_3) = (1, \theta_* b_2 e^{3\pi}, (1-\theta_*)b_3 e^{8\pi})$. They are fixed after this construction. Multiplication by $c = -1/(2\psi_a'(0)) > 0$ imposes the stated primitive endpoint; then $\psi_a'''(0) = \psi_a'(0) + 8\Phi_a'(0) = -1/2$. The raw third source derivative lies in $(-40.00, -39.98)$, so positive scaling does not restore source reflection.

The enclosure calculation uses validated integration on $[0, 1]$, together with the complete tail bound, for $k \le 3$,

$$\int_1^\infty u^k e^{u^2/4}e^{\lambda_n}\phi_n(u)\,du \le \lambda_n \exp\{\lambda_n - 4 - (\lambda_n/2)e^4\}.$$

Indeed $k\log u + u^2/4 + 9u \le (\lambda_n/2)e^{4u}$ for $u \ge 1$; its elementary endpoint constant is also enclosed. Centered jets and absolute moments extend the bounds across all of I. Fifty assertions at 160-bit precision and an independent replay support these finite claims; no sampled near-zero is substituted for root existence. □

The restriction to finite support is inessential. Add a sufficiently small positive multiple of $\sum_{n\ge 4}\phi_n$, and solve $\Phi_a'(0)=0$ for the third weight. After this elimination, the Jacobian of $(H_a, H_{a,x})$ with respect to the remaining weight parameter and x is a nonzero multiple of $B(x_*)H_{a,xx}(T,x_*)$. The implicit function theorem preserves the collision and endpoint, with every weight positive. Every finite ordinary and adjacent-square localizing matrix is then strictly positive definite. This continuation is qualitative: no numerical perturbation radius is asserted. All resulting weights and normalizations remain independent of physical heat time.

On the control's full stationary predecessor, $\mathcal{L}_0/E = -1/(T-t)+O(1)$; its short admissible chords also become negative. Thus the relaxed assumptions cannot supply the required uniform signed estimates. The control retains periodicity in z and period two in τ, and obeys Jacobi heat, but fails the full elliptic multiplier. Its evenness in the spectral variable x does not imply evenness of its source in u.

9.2 Complete reflection fixes the coefficients

Theorem 9.2 (Weighted-comb reflection rigidity)**.** *Let $(a_n)_{n\in\mathbb{Z}}$ be an even complex sequence with $|a_n|\le C(1+|n|)^M$ for some $C>0$ and finite $M\ge 0$. Define $F_a(q)=\sum_{n\in\mathbb{Z}} a_n e^{-\pi n^2 q}$ for $q>0$. If*

$$F_a(q) = q^{-1/2}F_a(1/q) \qquad (q>0), \tag{9.2}$$

then $a_n = a_0$ for every integer n. No coefficient positivity is needed.

Proof. Use $\widehat{f}(\xi)=\int f(x)e^{-2\pi i x\xi}\,dx$ and the even tempered comb $\mu=\sum a_n\delta_n$. For $g_q(x)=e^{-\pi q x^2}$, $\widehat{g}_q = q^{-1/2}g_{1/q}$, so (9.2) gives $\langle \nu, g_q\rangle = 0$, where $\nu=\mu-\widehat{\mu}$. Differentiation at $q=1$ in the Schwartz topology annihilates every even Gaussian moment of ν; odd moments vanish by evenness. The function

$$\mathcal{G}(z) = \langle \nu, e^{-\pi x^2+zx}\rangle$$

is entire, since the test functions and their parameter derivatives have uniform Schwartz seminorm bounds on compact z-sets. Every derivative at zero vanishes, hence $\mathcal{G}=0$. On the imaginary axis this represents the Fourier transform of $e^{-\pi x^2}\nu$; Fourier injectivity and local division by the nowhere-zero Gaussian give $\nu=0$. Thus $\mu=\widehat{\mu}$. Integer support makes $\widehat{\mu}$ period one as a distribution. Therefore μ is period one, and tests supported near successive integers give $a_{n+1}=a_n$. This proves the theorem; [15] records the full scope and defect identity. □

Corollary 9.3 (Source reflection and the primitive endpoint)**.** *In the class of Theorem 9.2, take $a_0=1$ and set, on the entire real u axis,*

$$P_a(u) = e^u F_a(e^{4u}) = e^u + 2\psi_a(u), \qquad \Phi_a = (P_a''-P_a)/16.$$

If $\Phi_a(-u)=\Phi_a(u)$ and $\psi_a'(0)=-1/2$, then $a_n=1$ for all n.

Proof. Local uniform convergence makes these functions real analytic. The difference $R(u)=P_a(u)-P_a(-u)$ satisfies $R''-R=0$, $R(0)=0$, and $R'(0)=2[1+2\psi_a'(0)]=0$. Hence P_a is even, which is exactly (9.2); apply the theorem. □

Together with the normalized primitive endpoint, the complete odd source endpoint hierarchy likewise gives reflection by real analyticity. Proposition 9.1 retains only its first condition. Alternatively, imposing the full elliptic law on $\sum_n a_n e^{\pi i n^2\tau+2\pi i n z}$ directly forces $a_{n+1}=a_n$ by reindexing. These relations supply coefficient coupling that support-only positivity omits. They identify the theta input, but an additional argument must still transfer that coupling to a favorable sign of the physical chord or stationary observation.

9.3 A normalized coefficient recurrence

The projected elliptic law supplies a convenient arithmetic input for the adjoint calculation below. Its normalization is essential: the projected law alone retains an affine ambiguity in the square coefficients.

Proposition 9.4 (The normalized recurrence core)**.** *Let $(a_n)_{n\ge 0}$ have polynomial growth, extend it evenly to $\mathbb{Z}$, and put*

$$\beta_n = (4\pi n^2 - 1)e^{-\pi n^2}, \qquad \Delta a_n = a_{n-1} - 2a_n + a_{n+1}.$$

Then the conditions

$$a_0 = 1, \qquad \Delta a_n = 0 \quad (n \ge 1), \qquad \sum_{n\ge 1} \beta_n a_n = \tfrac{1}{2} \tag{9.3}$$

force $a_n = 1$ for every n. No positivity assumption on a_n is needed. Moreover, the recurrence is exactly the vanishing of the complete one-shift defect after the physical source projection.

Proof. For the last assertion define

$$\Theta_a(z,\tau) = \sum_{n\in\mathbb{Z}} a_n e^{\pi i n^2\tau + 2\pi i n z}, \qquad (\mathsf{T}F)(z,\tau) = e^{\pi i\tau + 2\pi i z} F(z+\tau,\tau).$$

At $q = e^{4u} \ge 1$, use the complete ordinary source operator from (8.17),

$$\mathcal{S}_q F = e^u \left(\frac{q^2}{16\pi^2}\partial_z^4 + \frac{3q}{8\pi}\partial_z^2 \right) F(z, iq)\Big|_{z=0}.$$

Reindexing the shifted series and pairing opposite modes gives

$$\mathcal{S}_q(\mathsf{T}\Theta_a - \Theta_a) = \tfrac{1}{2}\sum_{n\ge 1} (\Delta a_n)\phi_n(u). \tag{9.4}$$

The zero mode is annihilated. If the right side vanishes for every $q \ge 1$, its least nonzero coefficient, if any, dominates as $q \to \infty$: its nonzero quadratic polynomial in q times $e^{-\pi n^2 q}$ decays more slowly than every later mode. Polynomial coefficient growth controls the remaining tail. Thus every Δa_n vanishes; the converse is immediate. This proves the stated equivalence.

The recurrence and $a_0 = 1$ give $a_n = 1 + bn$. The complete theta primitive has $\sum_{n\ge 1}\beta_n = 1/2$, so the endpoint in (9.3) implies

$$b\sum_{n\ge 1} n\beta_n = 0.$$

The last sum is strictly positive and finite, hence $b = 0$. The complete projected calculation and its primitive endpoints are recorded in [20]. □

The endpoint in (9.3) is $-\psi_a'(0) = 1/2$. It restores information lost by source projection; it does not turn coefficient rigidity into a signed transform estimate. In particular, the ramp $a_n = 1 + bn$ passes the projected recurrence before that normalization is imposed.

Remark 9.5 (Scope of finite endpoint restrictions). Positive weights on four square modes can satisfy $\Phi_a'(0) = \Phi_a'''(0) = 0$ and the normalized primitive endpoint while their physical field has a nondegenerate positive-time double zero [23]. Thus adding that second source endpoint does not replace the complete recurrence or reflection hypotheses. This control does not assert failure of every finite endpoint hierarchy.

9.4 Coupled addition and the complete starting factor

The arithmetic product must retain independent source points. Write $\Theta = \Theta_1$, $q = e^{4u}$, $r = e^{4v}$, with $u, v \geq 0$, and set $p = n + m$, $k = n - m$, $X = z + w$, $Y = z - w$. The inverse lattice map is $n = (p + k)/2$, $m = (p - k)/2$, with $p \equiv k \pmod 2$. Hence

$$\Theta(z, iq)\Theta(w, ir) = \sum_{p \equiv k \ (2)} e^{-\pi Q_{q,r}(p,k) + \pi i(pX + kY)}, \quad Q_{q,r} = \frac{q + r}{4}(p^2 + k^2) + \frac{q - r}{2}pk. \tag{9.5}$$

The quadratic form has eigenvalues $q/2, r/2 > 0$, so these series and their fixed jets converge normally on compact parameter sets.

For an explicit parity form, put $A = q + r$, $B = q - r$ and

$$\vartheta_\epsilon(Z, iA) = \sum_{a \in \mathbb{Z}} e^{-\pi A(a + \epsilon/2)^2 + 2\pi i(a + \epsilon/2)Z}, \qquad \epsilon = 0, 1.$$

Then the same identity is

$$\Theta(z, iq)\Theta(w, ir) = \exp\left(\frac{B}{2\pi}\partial_X \partial_Y\right) \sum_{\epsilon=0}^{1} \vartheta_\epsilon(X, iA)\vartheta_\epsilon(Y, iA). \tag{9.6}$$

The derivative expansion is normally convergent by the Gaussian margin $A - |B| = 2\min(q, r) > 0$. Its mixed derivative cannot be discarded: $q = r$ represents only the diagonal $u = v$ of the physical double integral.

Applying the two complete source operators to (9.5) gives

$$\Phi(u)\Phi(v) = \sum_{p \equiv k \ (2)} M_{p,k}(u, v), \qquad M_{p,k} = \tfrac{1}{4}\phi_{|(p+k)/2|}(u)\phi_{|(p-k)/2|}(v), \quad \phi_0 = 0. \tag{9.7}$$

This follows because each nonzero integer mode contributes $\phi_{|n|}/2$. The lines $p = k$ and $p = -k$ are annihilated by the source operators. Formula (9.7) retains positive source coefficients, but the physical transform still inserts its heat weights and oscillatory phases [21].

Proposition 9.6 (Endpoint cutoff in the coupled lattice)**.** *Let $h_{j,n} = \partial_x^j H_n$, $W = h_j(t, x)$ and $V_N = \sum_{m=1}^{N} h_{j,m}(t, y)$. The product WV_N, with its complete starting factor, uses precisely the coupled lattice strip*

$$p \equiv k \pmod 2, \qquad |p - k| \leq 2N, \tag{9.8}$$

with no restriction on $p + k$. More generally, bounded endpoint weights d_m give the mask $d_{|(p-k)/2|}$, independent of the starting index.

Proof. The endpoint index is $m = (p - k)/2$, while $n = (p + k)/2$ is unrestricted. Pairing the signs of the original integer indices gives the positive-index product $\sum_{n \geq 1} \sum_{m=1}^{N} h_{j,n}(t, x)h_{j,m}(t, y)$. The same argument gives the weighted mask; its zero-index value is irrelevant because $\phi_0 = 0$. Absolute source bounds justify the exchanges [22]. □

Remark 9.7 (Constraints on a signed use of the product). The source addition matrix is not a positive Gram kernel: its principal block on $\{0, 2\}$ has zero diagonal and strictly positive off-diagonal entries. The factor-preserving estimates used here recombine arithmetic parity sectors and the physical exchange sectors of a complete-start/cutoff-endpoint product before charging the error to the starting energy [21, 22, 24]. Full source reflection and generic source/Fourier Gram positivity also do not supply the missing sign: an explicit positive even source retains every odd primitive endpoint and nevertheless collides [24]. That source has no nonzero polynomial-growth integer-square expansion, so it does not contradict Theorem 9.2. The coupled identities fix the arithmetic bookkeeping; the following estimates must still control its complete signed physical observation.

10 Quantitative recurrence stability and endpoint currents

The coefficient recurrence also gives quantitative information. We prove stability for weighted perturbations and a finite endpoint approximation whose error retains the complete starting amplitude. Neither result supplies a theta sign; the unresolved lower boundary is identified explicitly.

10.1 Physical mode bounds and the imaginary anchor

Write $\Phi = \sum_{n\ge1} \phi_n$, where

$$\phi_n(u) = e^u(2\pi^2 n^4 e^{8u} - 3\pi n^2 e^{4u})e^{-\pi n^2 e^{4u}}, \quad H_n(t,x) = \int_0^\infty e^{tu^2}\phi_n(u)\cos(xu)\,du,$$

and put $h_{j,n} = \partial_x^j H_n$. Define

$$\beta_n = (4\pi n^2 - 1)e^{-\pi n^2}, \qquad \sum_{n\ge1}\beta_n = \tfrac12, \qquad m_\beta = \sum_{n\ge1} n\beta_n \ge \tfrac12. \tag{10.1}$$

The sum is the complete primitive endpoint $-\psi'(0)$.

Lemma 10.1 (Physical bound and endpoint evaluation)**.** *For every integer $j \ge 0$ and finite real $T_{\max}$, the constants*

$$M_j(T_{\max}) = \sup_{u\ge0} u^j e^{T_{\max}u^2 + 9u - (\pi/2)(e^{4u}-1)}, \qquad C_j(T_{\max}) = \frac{\pi M_j(T_{\max})}{4\pi - 1}$$

are finite, and

$$|h_{j,n}(t,x)| \le C_j(T_{\max})\beta_n \qquad (t \le T_{\max},\ x\in\mathbb{R}). \tag{10.2}$$

Moreover $\beta_n = 8H_n(0,i)$. Thus, for $\sum \beta_n|a_n| < \infty$, the mode sum $H_a = \sum a_n H_n$ and its real spatial jets converge absolutely, and $\sum \beta_n a_n = 1/2$ is exactly $H_a(0,i) = 1/16$.

Proof. Positivity of ϕ_n and its explicit formula give

$$\frac{\phi_n(u)}{\beta_n} \le \frac{2\pi^2}{4\pi-1} n^2 e^{9u - \pi n^2(e^{4u}-1)}.$$

Use $e^{4u} - 1 \ge 4u$ and $n^2 - 1/2 \ge n^2/2$, then integrate $n^2 e^{-2\pi n^2 u}$, to obtain (10.2). For $p_n(u) = e^u e^{-\pi n^2 e^{4u}}$, $p_n'' - p_n = 8\phi_n$ and $p_n'(0) = -\beta_n$. Integration against $\cosh u$ has zero upper boundary and gives $8\int_0^\infty \phi_n(u)\cosh u\,du = \beta_n$. Absolute summability justifies the stated mode sums and evaluation. □

The weighted class in this lemma need not have a smooth source density at zero. Its observations are defined by the convergent mode sums; polynomial growth of the coefficients is not required.

10.2 An exact normalized defect identity

For $a_0 = 1$ and $\sum \beta_n|a_n| < \infty$, set

$$d_k = a_{k-1} - 2a_k + a_{k+1}, \qquad e_a = \sum_{n\ge1}\beta_n a_n - \tfrac12.$$

For a real or complex test sequence with $|f_n| \le C\beta_n$, define

$$F_a = \sum a_n f_n, \quad F_\theta = \sum f_n, \quad m_f = \sum n f_n, \qquad G_f(k) = \sum_{n>k} (n-k) f_n \quad (k \ge 0).$$

Use G_β for the same construction with $f = \beta$, so $G_\beta(0) = m_\beta$.

Theorem 10.2 (Normalized recurrence stability)**.** *Put*

$$\eta_f = \frac{m_f}{m_\beta}, \qquad D_k(f) = G_f(k) - \eta_f G_\beta(k).$$

Then all the following series converge absolutely:

$$F_a - F_\theta = \eta_f e_a + \sum_{k\ge 1} D_k(f) d_k, \tag{10.3}$$

$$|F_a - F_\theta| \le C \left(|e_a| + 2 \sum_{k\ge 1} G_\beta(k) |d_k| \right). \tag{10.4}$$

The coefficients satisfy $|\eta_f| \le C$ and $|D_k(f)| \le 2C G_\beta(k) \le 8C\beta_{k+1}$. Their exact lower traces are

$$\Delta D_k = f_k - \eta_f \beta_k \quad (k \ge 1), \qquad D_0 = 0, \qquad D_1 = -F_\theta + \eta_f/2. \tag{10.5}$$

Proof. Writing $b = a_1 - 1$, the finite-index recurrence solution is $a_n = 1 + nb + \sum_{k<n} (n-k) d_k$. Since $\beta_{n+1}/\beta_n < 1/2$, $G_\beta(k) \le 4\beta_{k+1}$. Expanding d_k into its three neighboring coefficients shows $\sum G_\beta(k)|d_k| < \infty$ under the stated weighted assumption. The recurrence may therefore be summed against both β_n and f_n, giving

$$e_a = b m_\beta + \sum G_\beta(k) d_k, \qquad F_a - F_\theta = b m_f + \sum G_f(k) d_k.$$

Eliminating b proves (10.3). The bounds follow from $|G_f(k)| \le C G_\beta(k)$. Finally $\Delta G_f(k) = f_k$, $G_f(0) = m_f$, and $G_f(1) = m_f - F_\theta$, proving (10.5). □

For physical jets the estimate is uniform in real x and $t \le T_{\max}$. Its coefficients use no division by H, E, or stationarity. Gaussian decay of the dual weights does not, without a uniform defect-tail bound, imply a uniform Gaussian remainder after truncating the sum. At theta, $d = e_a = 0$; the identity then retains the unknown observation in D_1, rather than estimating its sign.

10.3 A finite positive hierarchy and its sharp rate

Proposition 10.3 (Finite recurrence stability)**.** *Suppose $a_0 = 1$, $a_n \ge 0$, $\sum \beta_n a_n = 1/2$, and $d_k = 0$ for $1 \le k \le N$. Put $M = N + 1$ and*

$$R_M^\beta = \sum_{n>M} \beta_n, \qquad m_M = \sum_{n=1}^{M} n\beta_n.$$

Then

$$\sum_{n\ge 1} \beta_n |a_n - 1| \le 2R_M^\beta + \frac{2m_M}{M}, \qquad |F_a - F_\theta| \le 2C \left(R_M^\beta + \frac{m_M}{M} \right). \tag{10.6}$$

The first bound is the exact supremum over this class. The order M^{-1} also occurs for the physical observation $H_a(0,0)$. If $a_1 = 1$ is additionally imposed, both bounds improve respectively to $2R_M^\beta$ and $2CR_M^\beta$, where $R_M^\beta = O((M+1)^2 e^{-\pi(M+1)^2})$.

Proof. The recurrence gives $a_n = 1 + bn$ for $0 \le n \le M$. Positivity and normalization give $-1/M \le b \le R_M^\beta/m_M$ and tail mass $\sum_{n>M} \beta_n a_n = R_M^\beta - b m_M$. Hence the weighted distance is at most $2R_M^\beta + (|b| - b)m_M$, proving (10.6). If $a_1 = 1$, then $b = 0$, giving the improved estimate.

For sharpness, choose $K > M$, set $a_n = (1 - n/M)_+$ except at K, and set $a_K = (R_M^\beta + m_M/M)/\beta_K$. This finite nonnegative sequence has every required constraint. For sufficiently large K, its weighted distance is $2R_M^\beta + 2m_M/M - 2\beta_K$. For the physical assertion put $\omega_n = \int_0^\infty p_n(u)\,du$. Integration by parts and $e^{4u} \ge 1 + 4u$ give

$$H_n(0,0) = \frac{\beta_n - \omega_n}{8}, \qquad 0 < \frac{\omega_n}{\beta_n} \le (4\pi n^2 - 1)^{-2}.$$

The ramp-and-spike deviation consequently tends, as $K \to \infty$, to

$$\frac{\sum_{n\le M} n\omega_n + M\sum_{n>M}\omega_n}{8M} \sim \frac{1}{8M}\sum_{n\ge 1} n\omega_n > 0.$$

For each M, sufficiently large finite K realizes at least half this positive limit. Finally the Gaussian tail bound for R_M^β follows directly from the explicit β_n. □

These controls establish a stability rate, not a collision. Even the anchored Gaussian absolute error is not automatically an $o(\sqrt{E})$ error near an unknown event.

10.4 A finite current preserving the starting amplitude

For a sequence f with $\sum_{n\ge1} n|f_n| < \infty$, adjoin $f_0 = 0$ and define

$$(\mathsf{P}f)_0 = 0, \qquad (\mathsf{P}f)_n = \tfrac{1}{2}(f_{n-1} + f_{n+1}) \quad (n \ge 1).$$

The zeroth coordinate is reset to zero at every iteration. For an integer depth $L \ge 1$, put $J^{[L]} = -\frac{1}{2}\sum_{k=0}^{L-1} \mathsf{P}^k f$ and $J_0^{[L]} = 0$.

Proposition 10.4 (Absorption current and remainder)**.** *Let* $c_L = 2^{-L}\binom{L}{\lfloor L/2\rfloor}$ *and set*

$$p_L(k) = 2^{-L}\binom{L}{(L+k)/2}, \qquad s_L(m) = \sum_{k=1-m}^{m} p_L(k), \qquad q_L(m) = 1 - s_L(m),$$

with inadmissible binomial coefficients interpreted as zero. Then $q_L(m) = 0$ *for* $m > L$, $0 \le q_L, s_L \le 1$, *and*

$$\Delta J^{[L]} = f - \mathsf{P}^L f, \qquad \sum f_n = B_L(f) + R_L(f), \tag{10.7}$$

$$B_L(f) = -J_1^{[L]} = \sum_{m=1}^{L} q_L(m) f_m, \qquad R_L(f) = \sum_{m\ge1} s_L(m) f_m, \tag{10.8}$$

$$|R_L(f)| \le c_L \sum_{m\ge1} m|f_m| \le \sqrt{\frac{2}{L+1}} \sum_{m\ge1} m|f_m|. \tag{10.9}$$

The first moment $\sum n(\mathsf{P}^L f)_n = \sum n f_n$ *is conserved.*

Proof. On sequences with zero zeroth coordinate, $\Delta = 2(\mathsf{P} - I)$, so the current identity telescopes. Reflection at zero gives the nonnegative kernel $p_L(n-m) - p_L(n+m)$ for P^L. Summing over $n \geq 1$ gives $s_L(m)$. Its defining interval contains m admissible parity indices, so $s_L(m) \leq \min(1, mc_L)$. The elementary central-binomial bound gives $c_L \leq \sqrt{2/(L+1)}$. Finite iterates preserve weighted absolute summability. Thus the upper boundaries in $\sum_{n=1}^{Q} \Delta J_n = J_0 - J_1 + J_{Q+1} - J_Q$ and $\sum_{n=1}^{Q} n\Delta J_n = J_0 - (Q+1)J_Q + QJ_{Q+1}$ vanish as $Q \to \infty$, proving the remaining assertions. □

Apply this construction only at the endpoint, defining

$$W^{[L]}(t,y) = \sum_{m=1}^{L} q_L(m) h_{j,m}(t,y), \qquad W = h_j.$$

Lemma 10.1 and Proposition 10.4 give $|W - W^{[L]}| \leq c_L C_j(T_{\max}) m_\beta$ uniformly for real y and $t \leq T_{\max}$. For $j = 0, 1$ and $\delta > 0$, define

$$\mathcal{A}_{j,\delta}^{[L]} = W(t,x)^2 - e^{-\delta} W(t,x) W^{[L]}(t, x+\delta), \quad \mathcal{A}_{j,\delta} = W(t,x)^2 - e^{-\delta} W(t,x) W(t, x+\delta).$$

Because the starting field remains complete,

$$|\mathcal{A}_{j,\delta} - \mathcal{A}_{j,\delta}^{[L]}| \leq e^{-\delta} c_L C_j(T_{\max}) m_\beta \sqrt{E(t,x)}. \tag{10.10}$$

This identity requires no stationarity assumption. Any independent signed-core bound must use the complete condition $h_{j+1}(t,x) = 0$ and its required interval domain. The coefficient of $\sqrt{E}$ in this error tends to zero at fixed δ; it is also smaller than every fixed power of $\delta_L = 1/\log(L+2)$. The finite primitive endpoint is $\beta^{[L]} = \sum_{m=1}^{L} q_L(m)\beta_m < 1/2$, with exact deficit $R_L(\beta)$, and cannot be replaced by $1/2$.

Remark 10.5 (Scope of the negative results). The current gives an approximation, not its sign. For a fixed residual with $\sum n|r_n| < \infty$, cancelling its boundary-plus-normalization term with $\eta_L = -B_L(r)/R_L(\beta)$ leaves exactly $R_L(r - \eta_L \beta) = \sum r_n$. The unknown observation has moved to the remainder. A separate finite-range result rules out the specified exact pointwise endpoint reconstruction by finitely many neighboring square modes, polynomial source multipliers and finitely many spectral frequencies. Its precise hypotheses require a nonzero diagonal target mode beyond the current's range and exclude an independent source-divergence term. These scoped obstructions, proved in [25], do not exclude integrated identities, source-divergence corrections with complete boundary terms, or one-sided bilocal estimates.

11 Arithmetic boundary traces and signed observation

The recurrence experiment exposes a boundary obstruction that the physical geometry can identify more precisely. The conclusions below concern exact sequence spaces, finite linear relaxations, and conditional physical estimates; they do not establish a signed theta inequality. Details are recorded in [26].

Let $\mathcal{S}_+$ and $\mathcal{S}_0$ denote the real rapidly decreasing sequences on $n \geq 1$ and $n \geq 0$, respectively, and set

$$(\Delta J)_n = J_{n-1} - 2J_n + J_{n+1}, \qquad \mathcal{N} = \{J \in \mathcal{S}_0 : J_0 = J_1\}.$$

Proposition 11.1 (The recurrence boundary class)**.** *The map $\Delta : \mathcal{N} \to \mathcal{S}_+$ is injective and*

$$\operatorname{range}(\Delta|_{\mathcal{N}}) = \left\{ r \in \mathcal{S}_+ : \sum_{n \geq 1} r_n = 0 \right\}. \tag{11.1}$$

Its algebraic cokernel is one-dimensional, detected by total sum.

Proof. Telescoping gives $\sum_{n\ge1}\Delta J_n = J_0 - J_1$. For any $r \in \mathcal{S}_+$, the current

$$J_n = \sum_{m>n} (m-n) r_m \tag{11.2}$$

is rapidly decreasing, satisfies $\Delta J = r$, and has $J_0 - J_1 = \sum r_n$. Rapid decrease follows from arbitrarily high weighted tail bounds. A homogeneous solution is affine in n, hence is zero if rapidly decreasing. This proves the range assertion and injectivity. □

The primitive endpoint sequence $\beta_n = (4\pi n^2 - 1)e^{-\pi n^2}$ has $\sum \beta_n = 1/2$. Its class therefore spans this cokernel: uniquely,

$$r = \Delta J + \eta\beta, \qquad J \in \mathcal{N}, \qquad \eta = 2\sum r_n.$$

Choosing η from the unknown physical residual does not estimate that residual. An effective arithmetic construction must control the lower trace independently, while retaining every finite-cutoff boundary. Parameter-dependent estimates additionally require locally uniform weighted bounds; a pointwise sequence identity does not supply them.

Proposition 11.2 (Bounded descent to the physical observation quotient)**.** *Let V be a finite-dimensional real inner-product space describing a homogeneous linear relaxation of arithmetic and stationarity constraints. Let $\mathsf{J} : V \to \mathbb{R}^2$ be its physical first-jet map and $\ell : V \to \mathbb{R}$ an endpoint functional. There exists a finite C such that*

$$|\ell(c)| \le C\|\mathsf{J}c\| \quad (c \in V) \tag{11.3}$$

if and only if $\ker \mathsf{J} \subseteq \ker \ell$. *In orthonormal coordinates the least constant is $C_{\min} = \|\ell \mathsf{J}^\dagger\|$, where $\mathsf{J}^\dagger$ is the Moore–Penrose inverse.*

Proof. Necessity follows on kernel vectors. Under kernel inclusion, $b(\mathsf{J}c) = \ell(c)$ is a well-defined covector on $\operatorname{im}\mathsf{J}$, with least bound its operator norm. The row $\ell\mathsf{J}^\dagger$ represents that covector and vanishes on the orthogonal complement of the image. □

Affine normalization requires retaining the basepoint contribution separately. This proposition does not assert that actual theta coefficients range over every vector in a chosen relaxation. Moreover, pointwise descent does not imply local boundedness through degeneration: for

$$\mathsf{J}(\tau) = \operatorname{diag}(1, \tau^2), \qquad \ell(\tau) = (0, \tau),$$

the kernel condition holds even at $\tau = 0$, but $C_{\min} = |\tau|^{-1}$ for $\tau \neq 0$. This is an analytic matrix example, not a theta heat field. It makes a vanishing-order estimate necessary after the rank test.

For the actual field, put $F = (H, H_x)$, $\rho = \|F\| = \sqrt{E}$, $W = h_j$, and $S = h_{j+1}$, with $j = 0, 1$. If an endpoint observation V has a reconstruction

$$V = AH + BH_x + CS + R, \qquad |R| \le \varepsilon, \tag{11.4}$$

then full stationarity with $E > 0$ gives the necessary estimate

$$\|(A, B)\| \ge \frac{(|V| - \varepsilon)_+}{\sqrt{E}}. \tag{11.5}$$

At a multiplicity $2r+j$ predecessor, $\sqrt{E} \asymp s^r$. A sufficiently close fixed endpoint, or a sufficiently accurate fixed cutoff of that endpoint, has a nonzero limit v. If $\varepsilon < |v|/2$, the coefficients in (11.4) must therefore grow at least as s^{-r}. An independently bounded reconstruction would exclude the event; division or pseudoinversion alone does not prove its boundedness. Bounded nonsingular changes of first-jet frame preserve this obstruction. A singular frame moves it into the comparison with the physical norm.

The actual chord target is less restrictive than absolute endpoint reconstruction. For $\rho > 0$ put $\theta = F/\rho$ and let e_j select the H or H_x coordinate. With $\alpha = e^{-\delta}$,

$$\frac{W^2 - \alpha WV}{E} = (e_j \cdot \theta)^2 - \alpha \frac{(e_j \cdot \theta)V}{\rho}. \tag{11.6}$$

For fixed $\delta > 0$, a lower bound by $-KE$ with locally bounded $K \geq 0$ is equivalent to local boundedness of $[WV]_+/E$. A large endpoint opposite in sign to W is harmless. The manuscript's marked real Gauss map retains orientation; here one must additionally retain the radial vanishing order relative to $\sqrt{E}$.

Proposition 11.3 (A sufficient signed boundary-trace estimate)**.** *Suppose a complete arithmetic decomposition gives*

$$V = AH + BH_x + CS + B_\partial + R, \qquad |R| \leq \varepsilon.$$

On full stationary starts with $\rho > 0$, suppose

$$\frac{W}{\rho} B_\partial \leq C_0 \rho + \varepsilon_0, \tag{11.7}$$

where $C_0, \varepsilon, \varepsilon_0 \geq 0$. Then

$$WV \leq (|A| + |B| + C_0)E + (\varepsilon + \varepsilon_0)\sqrt{E}. \tag{11.8}$$

Proof. Multiply (11.7) by ρ, use $S = 0$, and bound $|WH|$, $|WH_x|$ by E and $|WR|$ by $\varepsilon\sqrt{E}$. □

If A, B, C_0 are locally bounded and the errors meet the prescribed fixed-length or shrinking-length requirements, this gives the mixed-error bound for the exponential chord on both branches. The complete trace may take the recurrence form $B_\partial = J_0 - J_1 + \eta/2$, with additional cutoff or source boundaries retained explicitly. Neither the cokernel description nor a change of geometry proves (11.7). The new geometric task is an independent oriented estimate for this arithmetic trace with the required radial order; the nonnegative metric and its curvature magnitude do not supply that estimate.

12 The remaining arithmetic-to-physical estimate

The most focused open target combines the complete coefficient recurrence with the signed physical boundary observation. For both $j = 0, 1$, put $W = h_j$ and retain full stationarity $h_{j+1}(t, x) = 0$. For source-prescribed endpoint approximations $V_N(t, x+\delta)$, seek

$$W(t,x)^2 - e^{-\delta} W(t,x) V_N(t, x+\delta) \geq -K_{\delta,N} E(t,x) - \varepsilon_{\delta,N} \sqrt{E(t,x)}. \tag{12.1}$$

For each chosen length and cutoff, the coefficient must stay locally bounded through a hypothetical event. The endpoint approximation error and $\varepsilon_{\delta,N}$ must tend to zero along a prescribed cutoff sequence. The independent-cutoff criterion fixes a sufficiently small length, then an accurate cutoff,

and finally approaches the event; it imposes no growth restriction on $K_{\delta,N}$ across cutoffs. The theta field has not been shown to satisfy this bound.

The original chord bound $\mathcal{C}_j \geq -KE^{\gamma}$, $\gamma > 1/2$, and the stationary and angular criteria remain alternative sufficient hypotheses with their own coverage requirements. For a family available only at shrinking lengths, a sufficient uniform condition is that the error decay faster than every fixed power of those lengths, covering arbitrary finite multiplicity. Polynomial cutoff error permits lengths such as $1/\log(N+2)$; it does not justify an arbitrary faster schedule.

12.1 A signed trace from a joint source identity

The exact recurrence identifies the input and its dual quantifies defects, but at theta those defects vanish and the unresolved physical observation survives in the lower boundary. The finite-current normalization experiment does not remove it. A different bounded experiment would seek a source-prescribed joint identity

$$r_n(u) = \Delta_n J_n(u) + \partial_u K_n(u) + \varrho_n(u),$$

retaining both lower boundaries and every cutoff correction after summation and integration. Source divergence and primitive endpoints already occur in the complete-source formulas. Their mere reappearance is not an additional sign mechanism.

The useful geometric target concerns the complete resulting trace B_∂. If

$$V = AH + BH_x + Ch_{j+1} + B_\partial + R, \qquad |R| \leq \varepsilon,$$

with locally bounded A, B, seek

$$\frac{W}{\sqrt{E}} B_\partial \leq C_0\sqrt{E} + \varepsilon_0 \qquad (E > 0,\ h_{j+1} = 0). \tag{12.2}$$

The resulting mixed chord bound retains favorable large boundary values with the opposite sign from W. Absolute reconstruction is stronger than necessary. A new metric, normalization coefficient or pseudoinverse must produce an independent estimate of this signed trace and its radial order; a change of representation alone does not.

12.2 Earlier unsuccessful constructions

The coupled-product Gram ansatz, finite elliptic projection, and primitive normalization correction have the specific limitations recorded above and in their notes. These conclusions do not exclude all arithmetic currents, integrated cancellations or one-sided estimates. The period-two contour identity in [13] retains its shifted ray and bottom edge, whose leading terms cancel. Separate absolute sector bounds lose that cancellation and have exponential cost in spectral height. The detailed asymptotics remain in the note. Other geometric and outer-bridge alternatives are assessed in [16, 17, 27]. None supplies the missing signed trace or extends a theorem's physical domain.

12.3 Validation and coverage

A proposed coefficient rule should be tested against the asymmetric heat-polynomial and weighted-square controls, using the complete stationary condition. The positive recurrence hierarchy supplies an additional quantitative test: an argument claiming Gaussian convergence from its unanchored finite constraints conflicts with the sharp ramp controls. Conversely, even Gaussian absolute proximity to theta does not establish the sign of theta's observation.

A theta pilot must evaluate the complete signed pairing, full stationary root, interval conditions, endpoints and tails with coefficients fixed before the calculation. A finite success motivates an analytic extension; it does not prove the required cutoff family or coverage of every positive-time finite-height event. The declared local uniformity, event coverage and imported global hypotheses in Remark 2.2 remain indispensable for an RH conclusion. A single constant uniform as $t \downarrow 0$ or $|x| \to \infty$ is not asserted or required by the fixed-length mixed-error criterion. The next-chat continuation is [28].

Acknowledgments and research status

This research manuscript was prepared for Edward Baker with substantial assistance from OpenAI language models. The coordinating session used `gpt-6-astra` with reasoning effort `ultra`; internal collaborators inherited that configuration. LLM assistance was used for drafting, derivations, literature checks, exact-arithmetic programs and internal reviews. The accompanying reviews distinguish analytic arguments, validated interval computations, and finite algebra checks. These internal checks are not external mathematical refereeing or formal verification. Edward B. Baker III's 2014 quaternion-form notes are acknowledged as exploratory motivation. The author and future reviewers retain responsibility for assessing the claims and their applicability.

A Verification scope and provenance

The verification package distinguishes analytic arguments from finite calculations. Seven exact checkers cover the geometry, Chern/ASD identities, complete signed bridge and heat profiles, with 1,230 overlapping assertions. Two further exact checkers add 5,667 assertions for the chord controls and reflection algebra. The weighted-square certificate and contour checker add 123 validated or mixed assertions using Arb/Acb with explicit complete tails. These totals are not counts of distinct theorems. The reflection-rigidity and arbitrary-multiplicity arguments are analytic proofs, not consequences of finite samples.

The chord replay uses 24 asymmetric or sign-reversed heat germs, 144 full stationary roots, and 1,152 zero-free intervals at three scales. Early intervals need not show the eventual negative sign; the asymptotic theorem is not replaced by universal finite-step negativity. The weighted-square calculation certifies a non-theta collision. The contour samples check complete identities and are not stationary theta sign certificates.

The later coefficient-core and endpoint package records 2,136 exact and 26 mixed or validated checks; the coupled-product package records 3,278 exact and 38 mixed or validated checks. The recurrence checker adds 6,892 exact rational assertions, independently replayed. Its rational surrogate weights test finite current, defect and hierarchy algebra; they are not actual-theta numerical sign data. The physical beta bound, sharp hierarchy rate, sequence-space range and quotient-norm statements are analytic proofs. These packages are indexed in the coefficient-core manifest, coupled-product manifest, and recurrence research manifest.

The reproducible scripts, output receipts and source hashes are indexed in the connection and profile manifest and arithmetic, contour and chord manifest. The current integration review records the manuscript-specific proof and layout checks; the build record identifies this source and its exported PDF. Older manifests describe their own research checkpoints, including the then-unchanged initial manuscript. They are not silently rewritten as current build records.

The foundational material comes from Investigation 21 [1] and its Notes 56–63. Principal geometric arguments are restated, while the global theta-to-RH bridge retains its imported status. The

current integration adds exponential-average transfer, normalized coefficient coupling, quantitative recurrence stability, finite endpoint currents and the signed boundary formulation. Negative findings are summarized with their precise scopes and links to detailed notes; less developed proposals remain in [16, 27]. Local research links are relative to the repository reading copy in `output/pdf`; the source remains the stable editable manuscript, with milestones indexed in `DRAFT_HISTORY.md` and no new snapshot folders.

References

[1] E. Baker (research project), *Theta correlations, Laguerre signs, and hypercomplex jets of the theta heat flow*, Investigation 21, stable research manuscript, 15 October 2026; drafted for Edward Baker with substantial LLM assistance. The accompanying research continuation includes Notes 56–63.

[2] E. B. Baker III, *Quaternion-form working notes with applications to Yang–Mills theory*, unpublished exploratory drafts, 21 January and 14 February 2014. The January version specifies $SU(2)$ in its title. Both versions are retained in Investigation 21's source documents.

[3] E. Baker (research project), *Positive primitive reconstruction can coexist with a positive time collision*, Investigation 21, Note 56, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[4] D. Joyce, *Lectures on Calabi–Yau and special Lagrangian geometry*, arXiv:math/0108088 (2001), revised 2002. An expanded version appears in M. Gross, D. Huybrechts and D. Joyce, *Calabi–Yau Manifolds and Related Geometries*, Springer, 2003.

[5] J. D. Lotay, *Calibrated submanifolds*, arXiv:1810.08709 (2018).

[6] J. E. Andersen and J. L. Blaavand, *Asymptotics of Toeplitz operators and applications in TQFT*, Travaux Mathématiques **19** (2011), 167–201; arXiv:1106.1564.

[7] M. Gross and P. M. H. Wilson, *Large complex structure limits of K3 surfaces*, Journal of Differential Geometry **55** (2000), no. 3, 475–546; doi:10.4310/jdg/1090341262; arXiv:math/0008018.

[8] D. Joyce, *The exceptional holonomy groups and calibrated geometry*, in S. Akbulut, T. Önder, and R. J. Stern (eds.), *Proceedings of the Gökova Geometry-Topology Conference 2005*, International Press, Somerville, MA, 2006, 110–139; arXiv:math/0406011 (2004).

[9] E. Baker (research project), *Inner turning profile and physical energy on both stationary branches*, Investigation 22, Signed turning Note 01, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[10] E. Baker (research project), *Central-energy chord and cutoff transfer*, Investigation 22, Signed turning Note 02, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[11] E. Baker (research project), *ASD metric freedom and physical restriction*, Investigation 22, Theta connections Note 02, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[12] E. Baker (research project), *Complete signed correlation bridge*, Investigation 22, Theta connections Note 03, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[13] E. Baker (research project), *Period-two contour packet and sector obstruction*, Investigation 22, Theta connections Note 04, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[14] E. Baker (research project), *Square-support endpoint collision obstruction*, Investigation 22, Control families Note 01, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[15] E. Baker (research project), *Weighted-comb reflection rigidity*, Investigation 22, Control families Note 02, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[16] E. Baker (research project), *Chord criterion, arithmetic bridge and next research steps*, Investigation 22, Research overview 02, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[17] E. Baker (research project), *Prioritized signed-bound research paths*, Investigation 22, Research coordination Note 03, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[18] E. Baker (research project), *Factor-preserving and averaged chord targets*, Investigation 22, Signed turning Note 03, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[19] E. Baker (research project), *Exponential average and shrinking chord targets*, Investigation 22, Signed turning Note 04, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[20] E. Baker (research project), *Elliptic shift core and projection obstruction*, Investigation 22, Theta connections Note 05, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[21] E. Baker (research project), *Coupled theta product and parity sectors*, Investigation 22, Theta connections Note 06, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[22] E. Baker (research project), *Parity strip cutoff and full starting factor*, Investigation 22, Theta connections Note 07, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[23] E. Baker (research project), *Two-endpoint square collision*, Investigation 22, Control families Note 03, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[24] E. Baker (research project), *Reflection-positive source collision and mixed sectors*, Investigation 22, Control families Note 04, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[25] E. Baker (research project), *Adjoint recurrence endpoint experiment and quantitative stability*, Investigation 22, Theta connections Note 08, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[26] E. Baker (research project), *Arithmetic boundary traces and signed physical observation*, Investigation 22, Physical jet geometry Note 01, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[27] E. Baker (research project), *Research paths after the coupled-product experiment*, Investigation 22, Research coordination Note 07, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.

[28] E. Baker (research project), *Recurrence manuscript integration and next-chat continuation*, Investigation 22, Research coordination Note 09, 15 October 2026; prepared for Edward Baker with substantial LLM assistance.